University of Pisa

EXplainable Artificial Intelligence Course:

Project Report, 3 March 2024

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Introduction

Breast cancer is a disease in which cells in the breast grow out of control. This disease is a common cause of death for women in the world. For this reason, early detection, diagnosis and treatment are important to decrease the mortality rate. Early detection requires an accurate classification method of tumors [1]. A tumor (or neoplasm) is an abnormal mass of cells in the body, which is caused by cells dividing more than normal or not dying when they should. There are two types of tumors: benign and malignant tumors. Benign tumors stay in their location without invading other sites of the body, tend to grow slowly and have distinct borders. On the opposite, cells in a malignant tumor grow uncontrollably and spread locally and/or to distant sites. Malignant tumors are cancerous [2].

Existing tumor classification methods are based on the analysis of texture and morphological features of breast ultrasound images. Traditionally, well-trained and experienced medical doctors (radiologists) used to conduct the analysis manually by checking texture and morphological features in images. However, there are a few limitations in this approach. A recent advance in data collection and artificial intelligence allows to train tumor classifiers based on texture and morphological features. However, these classifiers are black-box models, meaning such models do not offer any explanation on how images classified [3]. This is a risky situation, especially when it concerns human health. Moreover, the application of black-box models raises legal and ethical issues.

Our report tries to address the explainability issue by producing and examining explanations of predictions by the trained model. Firstly, we got familiar with a tabular dataset describing breast tumor images and trained a simple neural network with the data. We also explored model results in terms of accuracy, precision, recall and F-1 score. Then, we chose 10 instances from the dataset, where in 3 of them the class was mispredicted, in 3 instances the model had low confidence in prediction, while in other 4 instances the model had high confidence. Next, we applied post-hoc explainability techniques on each instance, using three main models (LIME, LORE and SHARP) covered during the course. In the end, we trained interpretable models (a logistic regression model and a decision tree) on the same dataset and produced explanations for 10 instances chosen. We also evaluated the model in terms of fidelity. Our projects are based on methods offered in [3].

The Dataset and the AI Model

Breast Cancer Wisconsin (Diagnostic) Data Set is one of the commonly used tumor data sets, which is available in the well-known machine learning data repository of UCI [4]. The dataset is represented as a tabular description of breast tumor images. The features describe the characteristics of the cell nuclei represented in the original images. Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. Each instance has an ID number, diagnosis (M = malignant, B = benign), and

following real-valued features: radius, texture, perimeter, area, smoothness (local variation in radius lengths), compactness (`perimeter^2 / area - 1.0`), concavity (severity of concave portions of the contour), concave points (number of concave portions of the contour), symmetry and fractal dimension ("coastline approximation" - 1). After adding mean value, standard error and maximum value of each feature, there are 32 features. There are 569 data instances (357 benign, 212 malignant diagnoses) and no missing attribute value.

We used a simple neural network model provided for the explainability project, the goal of the model is to predict if the cell indicates a benign or malignant tumor. First, the data is converted into PyTorch sensors, then a simple model architecture is defined in PyTorch. Next, the model is trained and tested given hyperparameters. The model has 92.98% accuracy, 84.31% precision, 100% recall, and F-1 score of 91.49%. These results look significantly high for a small data set. However, we would like to see how the model could predict the class correctly, and when it failed to do so.

Producing Explanations for the AI Model

We selected 10 instances from the dataset, spanning across various categories, to observe the performance of the model under different conditions while utilizing different explainability techniques. To do so, we selected 3 instances with the misclassified output, 3 instances predicted with low confidence and 4 instances predicted with high confidence.

0.1 LIME

LIME (Local Interpretable Model-agnostic Explanations) is a novel explanation technique designed to provide interpretable and faithful explanations for the predictions of any classifier. Unlike many traditional explanation methods, LIME operates locally, focusing on explaining individual predictions rather than the overall behavior of the model. It achieves this by learning an interpretable model in the vicinity of a specific prediction, effectively capturing the local decision boundary.

At the heart of LIME lies the concept of perturbation: by generating small variations or perturbations around a given instance, LIME observes how these perturbations affect the model's predictions. By sampling and analyzing the model's response to these perturbed instances, LIME constructs a simplified, interpretable surrogate model that approximates the behavior of the original classifier within the local neighborhood of the input space. This surrogate model, often a linear model, serves as the basis for generating explanations.

One of the key strengths of LIME is its model-agnostic nature, allowing it to be applied to any classifier without requiring knowledge of the underlying model's architecture or parameters. This versatility makes LIME suitable for a wide range of machine learning tasks, including classification, regression, and even natural language processing.

Figure 1 illustrates a LIME explanation for instance one in the misclassification category. As is evident, on the left side of the figure, the production probabilities of the instance for class 0 are 11%, while for class 1, it has been predicted to be 89%. This misclassification occurs as the instance actually belongs to class 0. In the middle part of the figure, LIME highlights features in two colors: blue indicates factors influencing the instance to be classified as class 0, while orange features support class 1. Additionally, we can observe the contribution of each feature to the categorization of the instance, represented by the coefficients of the model trained to explain this instance.

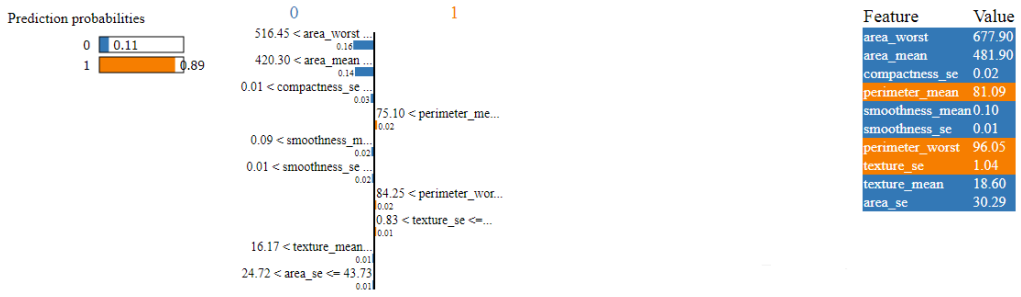


Figure 1: **Lime Explanation for The First Misclassification Instant.**

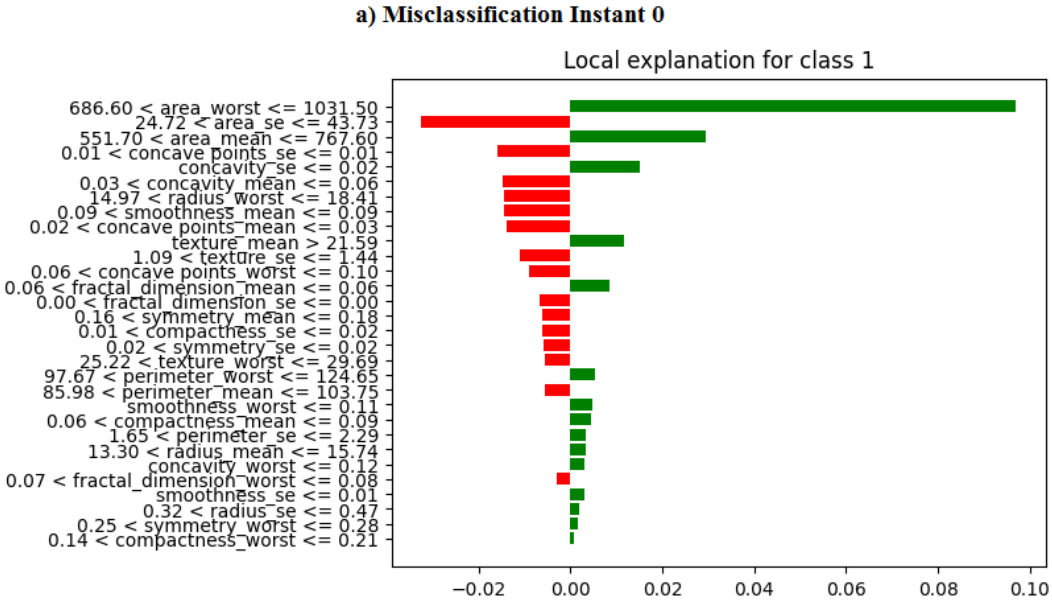


Figure 2: **Lime Bar chart Explanation for The First Misclassification Instant.**

Based on the model's analysis, the first instance has been classified as belonging to class 1, indicating a diagnosis of malignant cancer. Upon closer examination, it's apparent that the perimeter-mean feature exceeds 75.1 units, while both perimeter-worst and texture-se are above 84.25 and 0.83, respectively. These features collectively contribute to the model's prediction of malignancy for this instance. On the other hand, Area-worst and area-mean are lower than 686.6 and 481.9, respectively. these features supports instance to be classified as class 0 in the case that they can pass determined thresholds.

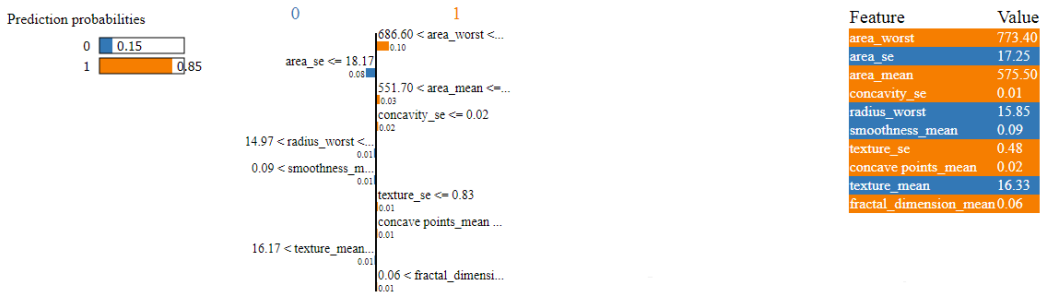


Figure 3: **Lime Explanation for The Second Misclassification Instant.**

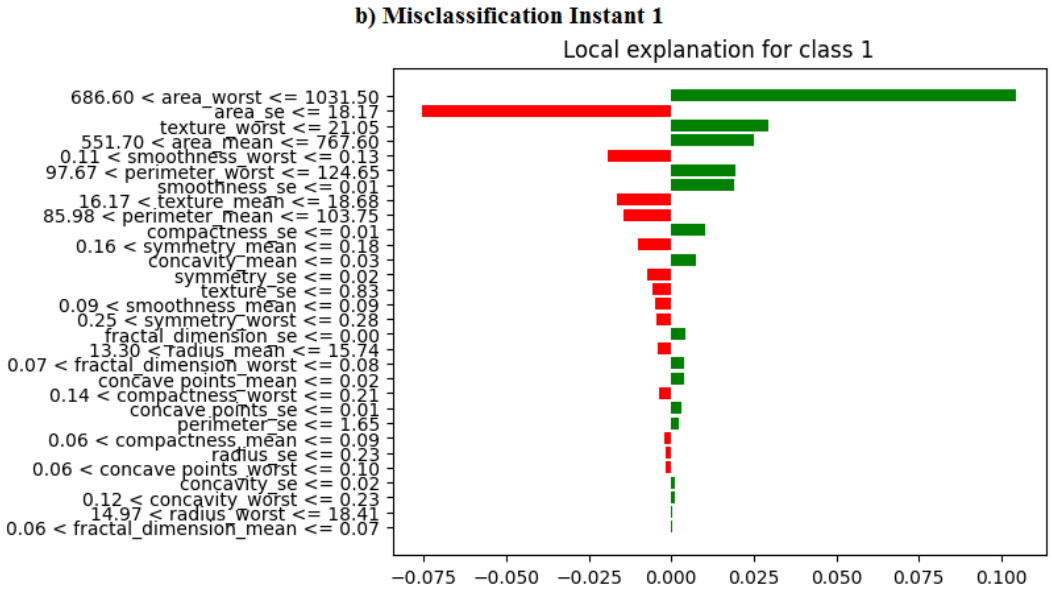


Figure 4: **Lime Bar chart Explanation for The Second Misclassification Instant.**

Furthermore, the second instance in our analysis has been misclassified as belonging to class 1. Upon examination of the prediction probabilities generated by the model, it becomes apparent that there is an 85% likelihood for this instance to be classified as class 1, with a remaining 15% probability of it being classified as class 0, or benign. In making this decision, features such as area-worst and area-mean emerge as pivotal factors, each surpassing critical thresholds of 686.6 and 551.7, respectively.

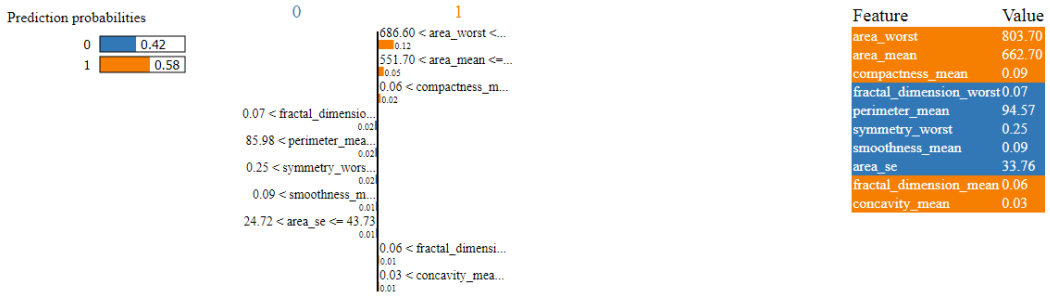


Figure 5: **Lime Explanation for our third Misclassification Instance.**

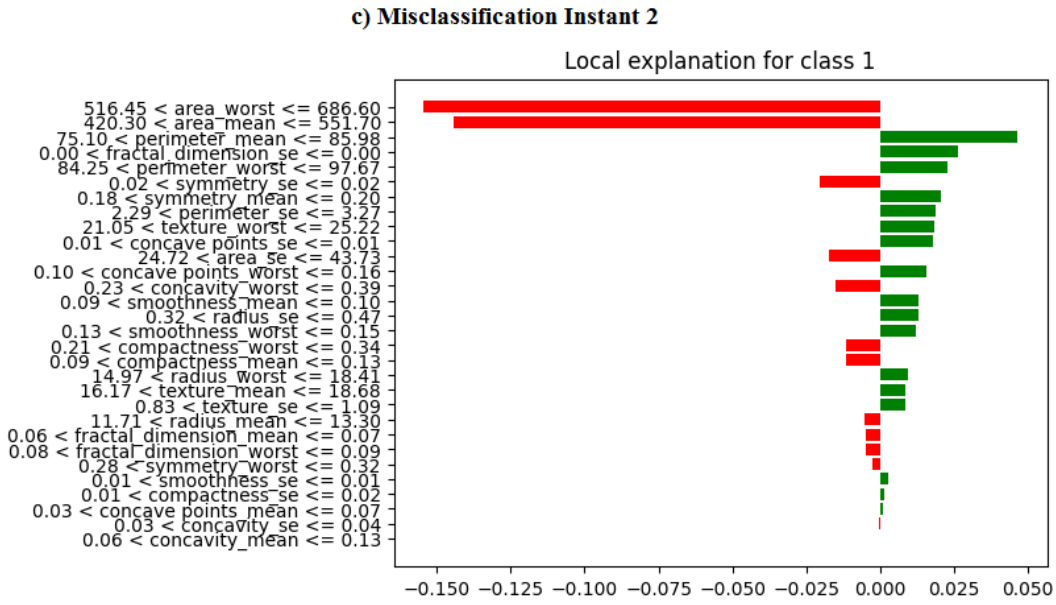


Figure 6: **Lime Bar chart Explanation for The Third Misclassification Instance.**

In the third instance of misclassification, the model's decision was influenced by a very marginal difference. According to the prediction probabilities generated by the model, there is a 58% likelihood that this particular cancer would be classified as malignant, while the probability of it being benign stands at 42%. Despite this close margin, certain features emerged as significant contributors to the model's decision-making process. Notably, features such as area-worst and area-mean played a predominant role. These orange-highlighted features likely influenced the model towards a classification of malignancy. On the other hand, features such as perimeter-mean and area-se appeared to support the classification of benign tumors.

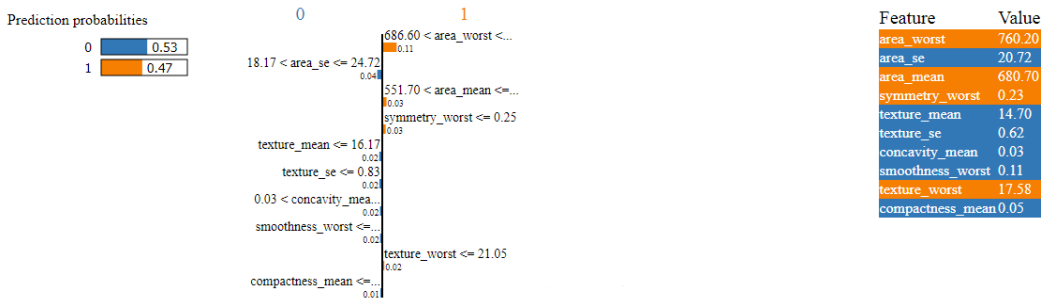


Figure 7: **Explanation for The First Low Confidence Prediction.**

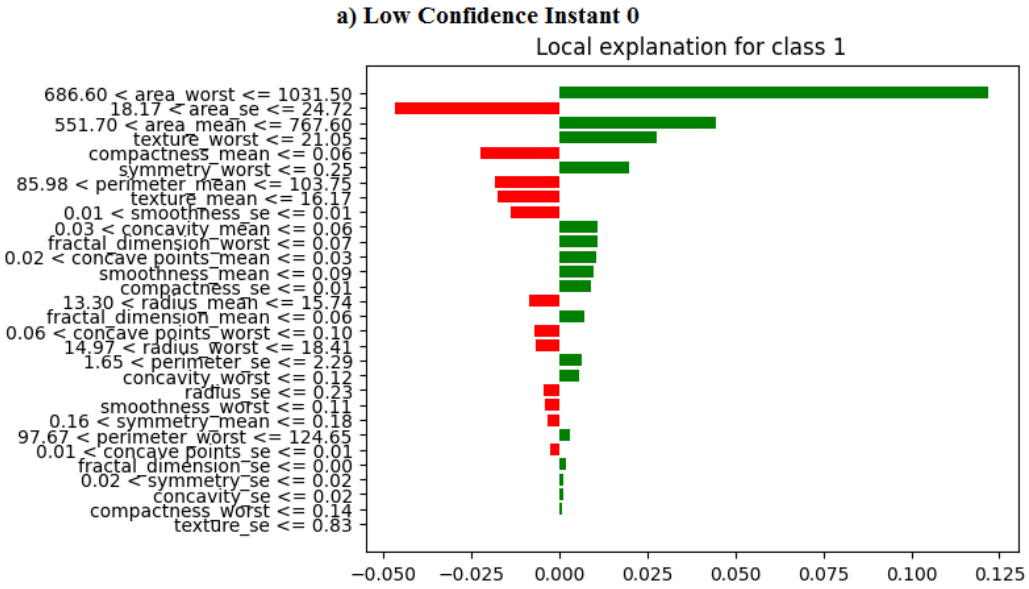


Figure 8: **Lime Bar chart Explanation for The First Low Confidence Prediction.**

The second major category among our instances is characterized by low confidence in the model's predictions. While the model's decision is ultimately correct, the difference between the probabilities assigned to the two classes is minimal. In the first instance, for example, the model predicted a 53% chance of the tumor being benign, with a slightly lower 47% probability of malignancy. Despite the closeness of these probabilities, the model's decision aligns with the ground truth. Upon closer examination of the features, it becomes apparent that blue-highlighted features predominantly support the classification of class 0, or benign tumors. Specifically, features such as area-se and texture-mean, with values of 20.72 and 14.7 respectively, play crucial roles in this decision.

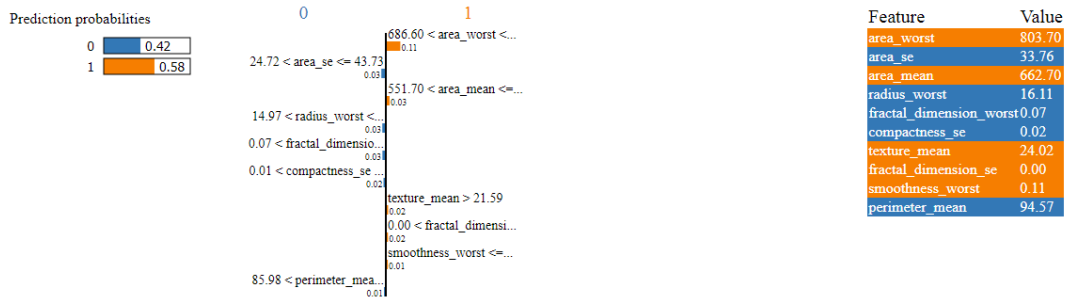


Figure 9: **Explanation for The Second Low Confidence Prediction.**

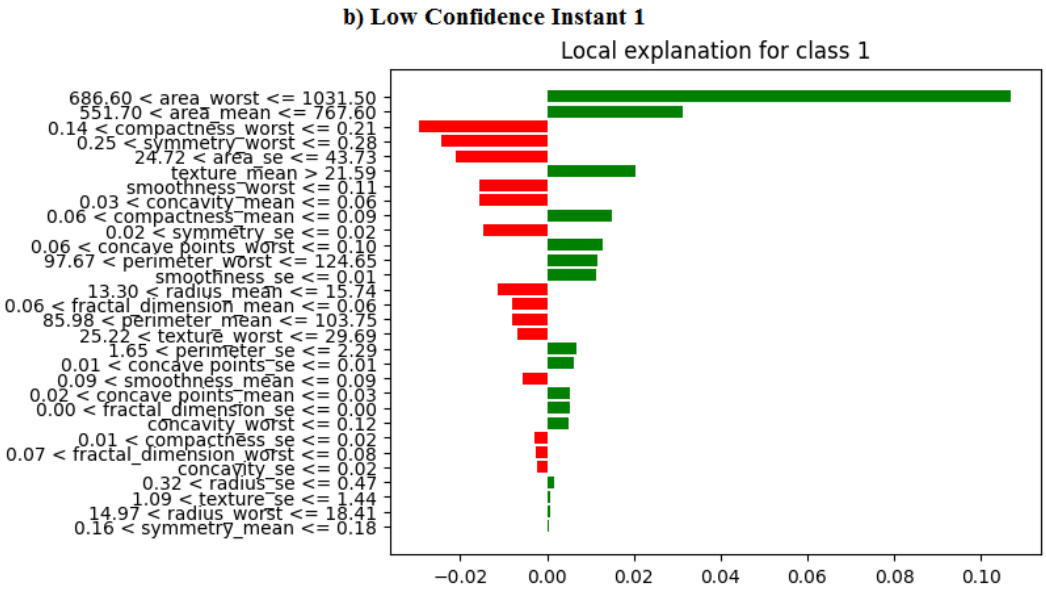


Figure 10: **Lime Bar chart Explanation for The Second Low Confidence Prediction.**

Moreover, in the second instance, the model assigned a probability of 58% for the tumor to belong to class 1. Here, the influence of orange-highlighted features is notable, indicating a significant contribution to the model's decision. Particularly, features such as area-worst and area-mean are highlighted, surpassing critical thresholds of 686.6 and 551.7 respectively. These values signify characteristics often associated with malignant tumors, such as irregular shape and increased size.

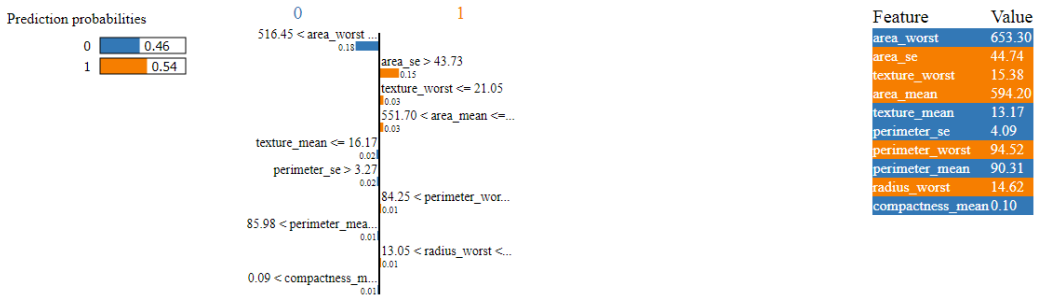


Figure 11: **Explanation for The Third Low Confidence Prediction.**

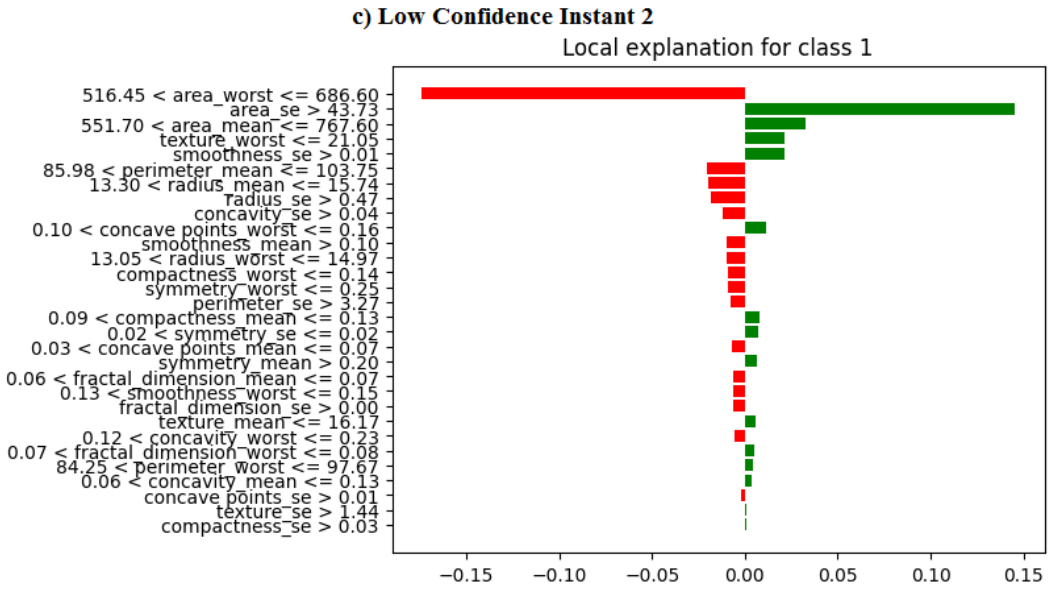


Figure 12: **Lime Bar chart Explanation for The Third Low Confidence Prediction.**

Finally, in third patient with model's low confidence prediction there is 54% chance of being from class 1 and 46% class 0. Among features of this instance area-se, texture-worst and area-mean have significant impact with values of 44.74, 15.38 and 594.2, respectively.

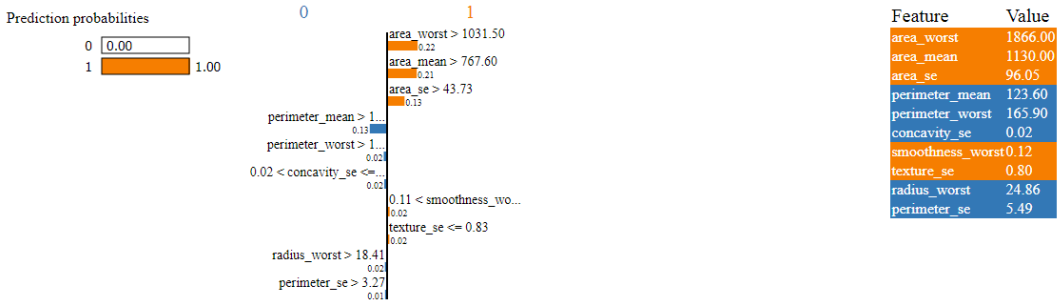


Figure 13: **Explanation for The First High Confidence Prediction.**

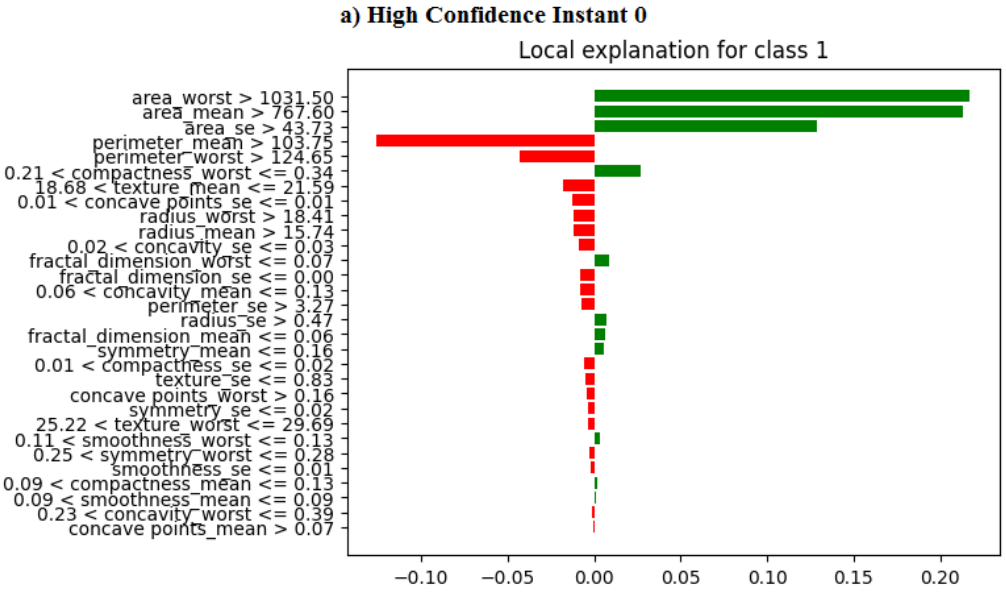


Figure 14: **Lime Bar chart Explanation for The First High Confidence Prediction.**

Our third category of instances falls under the classification of 'High confidence', where the model's predictions align strongly with the actual classes, accompanied by robust probabilities. As depicted in Figure 13, the model assigns a probability of 100% for the first instance to belong to class 1. Examining the middle part of the figure reveals that features such as area-worst, area-mean, and area-se play pivotal roles in driving this decision. Notably, area-worst stands out as the highest supporter of class 1, highlighted in orange, and surpasses a threshold value of 1031.5. This feature's prominence underscores its significance in distinguishing malignant tumors, as it often reflects the irregular shape and increased size characteristic of such tumors.

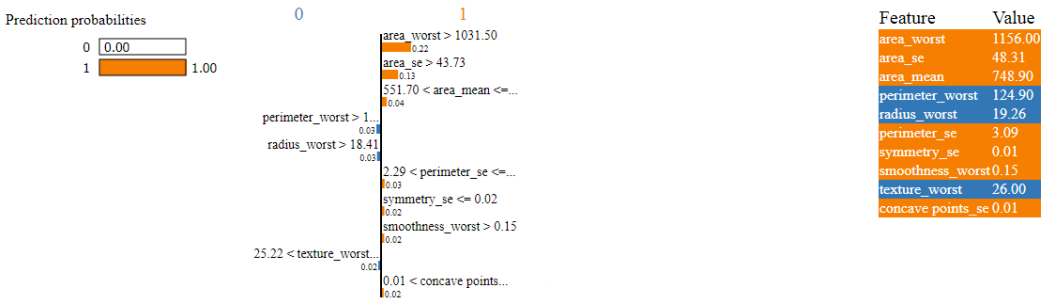


Figure 15: **Explanation for The Second High Confidence Prediction.**

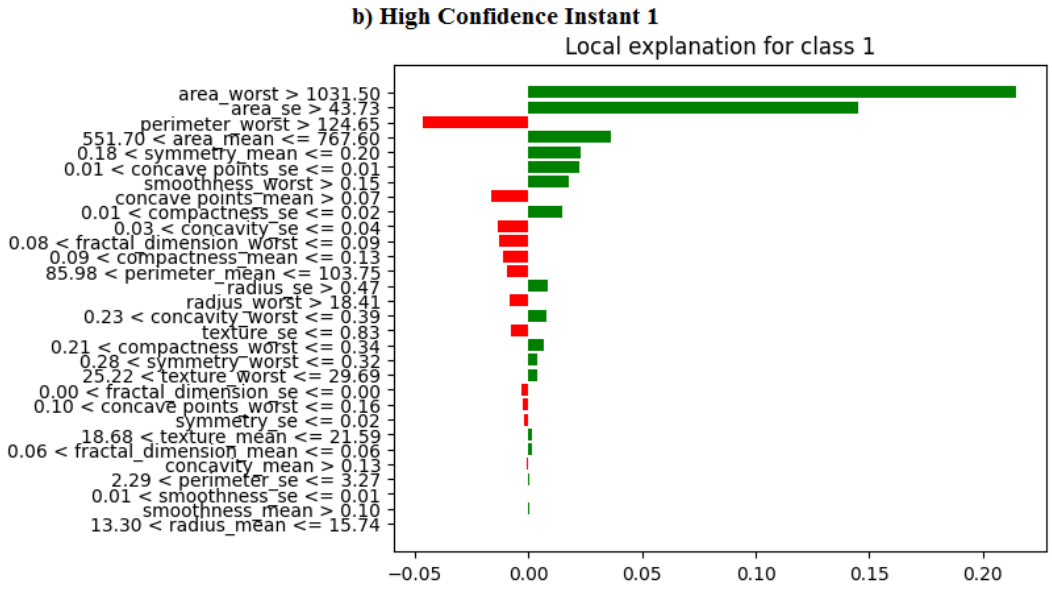


Figure 16: **Lime Bar chart Explanation for The Second High Confidence Prediction.**

The proposed model confidently predicted the second instance to be malignant, falling within the 'High confidence' category. As depicted in Figure 15, key features such as area-worst, area-se, and area-mean exhibit values of 1156, 48.31, and 748.9 respectively, significantly contributing to the model's decision. These features play a crucial role in classifying the instance as belonging to class 1. Notably, the elevated values of area-worst and area-mean, accompanied by the presence of area-se, suggest characteristics commonly associated with malignant tumors.

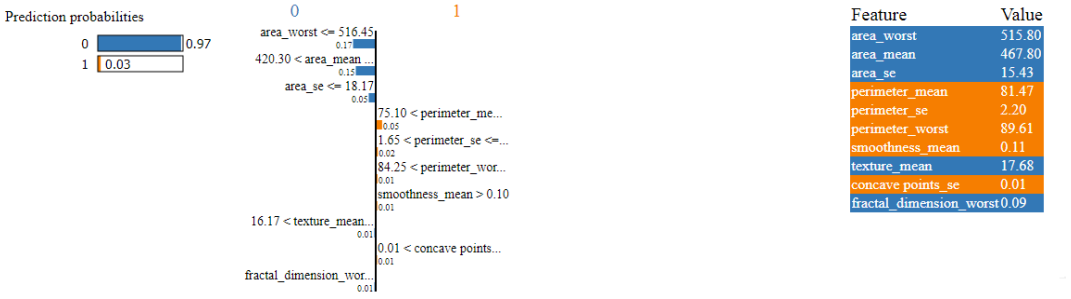


Figure 17: **Explanation for The Third High Confidence Prediction.**

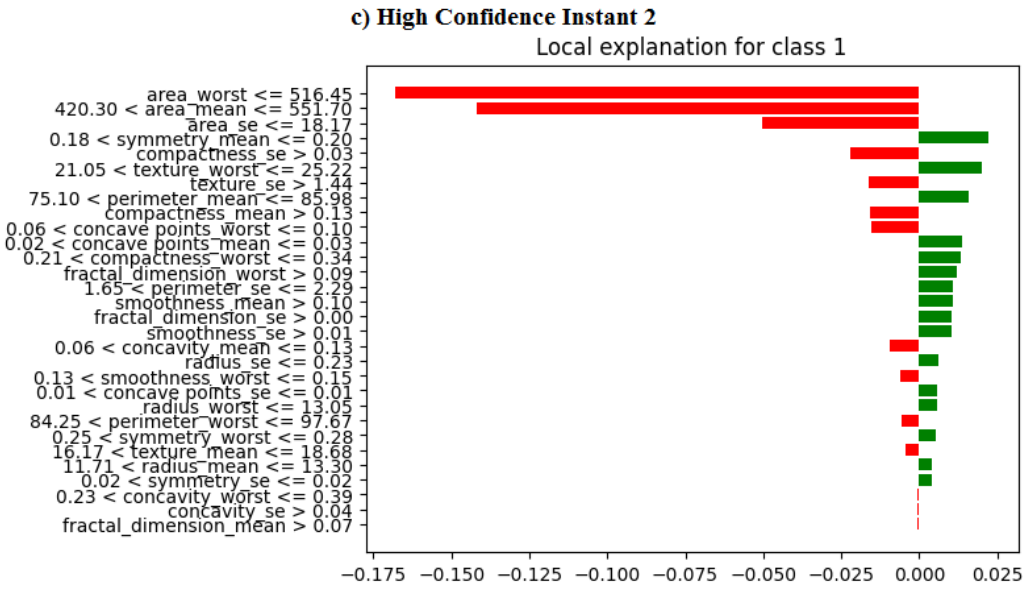


Figure 18: **Explanation for The Fourth High Confidence Prediction.**

Lastly, the model classifies the third instance as belonging to class 0. In Figure 17 of the LIME explanation, we observe that features such as area-worst, area-mean, and area-se contribute to the classification of class 0 with values of 515.8, 467.8, and 15.43 respectively. Additionally, Figure 18 illustrates the thresholds for these features, which are determined to be 516.45, 551.7, and 18.17 respectively. These features collectively support the classification of the instance as class 0. Notably, the values of these features fall within the specified thresholds.

0.2 LORE

LORE (Local Rule-Based Explanations of Black Box Decision Systems) is an explainable artificial intelligence (XAI) technique designed to provide local, interpretable explanations for the decisions made by complex, black-box machine learning models. It aims to shed light on the decision-making process of these models, particularly in situations where traditional methods of interpretation may not be feasible due to the complexity or opacity of the model.

At its core, LORE operates by generating simplified, rule-based explanations that approximate the behavior of the black-box model in the local vicinity of a specific instance. Instead of attempting to explain the entire global behavior of the model, LORE focuses on generating explanations tailored to individual predictions, providing insights into why a particular decision was made for a given input.

The key feature of LORE is its rule-based approach to explanation generation. It extracts local decision rules from the black-box model, which are then used to explain the model's predictions in a human-understandable format. These rules typically take the form of logical statements or conditions based on the input features, making them easily interpretable by users.

LORE can be applied to a wide range of machine learning models, including decision trees, random forests, support vector machines, and neural networks, without requiring access to the internal workings of these models. This flexibility makes LORE suitable for various tasks, such as classification, regression, and anomaly detection.

One of the advantages of LORE is its ability to provide transparent and actionable insights into the decision-making process of black-box models, helping users understand

the factors driving specific predictions. By generating interpretable rules at a local level, LORE enables users to identify important features and their interactions, detect biases or anomalies, and validate the model's behavior against domain knowledge or regulations.

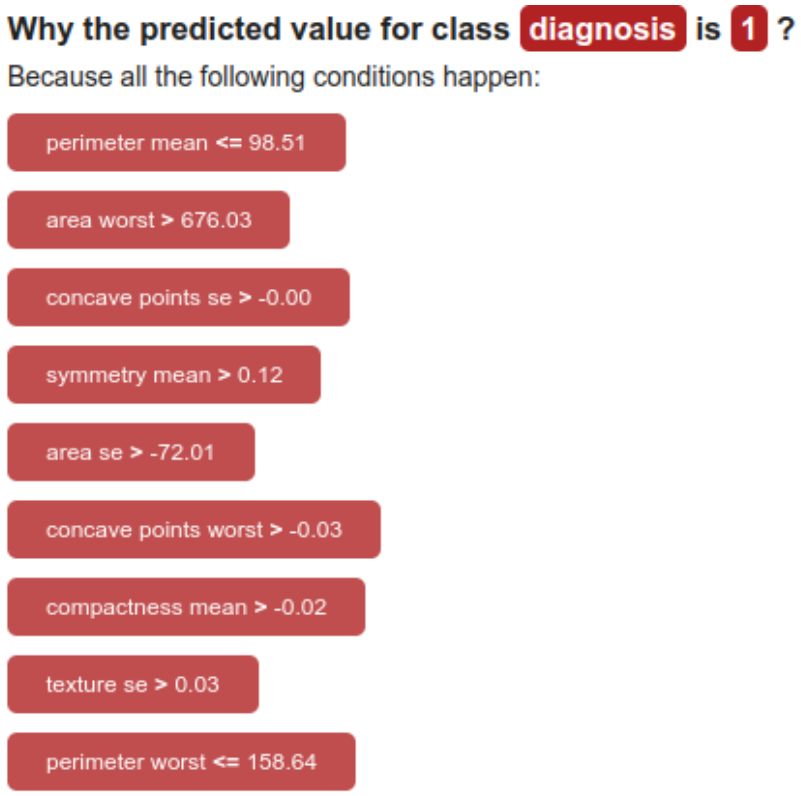


Figure 19: **Lore explanation of rules contributed in classification for third instance**

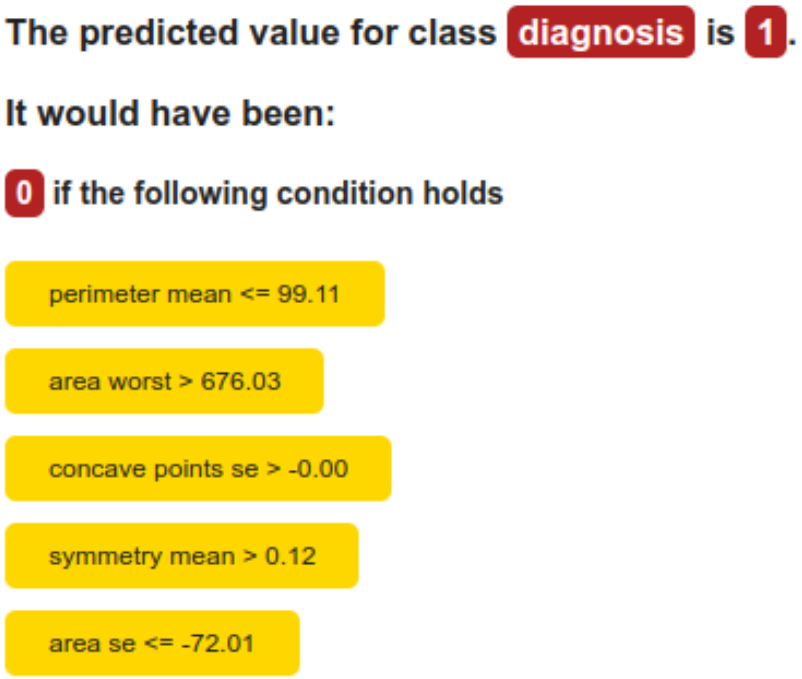


Figure 20: **Plot of counter factual Rules for third instance**

In the first instance, a misclassification occurred where the model classified the instance as belonging to class 1. Figure 19 provides insights into the features that predominantly influenced this decision. According to the LORE explanation, the perimeter-mean feature was observed to be under 98.51, while the area-worst feature exceeded 676.03. These specific values of the features played crucial roles in driving the classification towards class 1.

Additionally, counterfactual rules are presented to illustrate which features would have supported the classification of the instance into class 0. These rules highlight the conditions under which the instance would have been classified differently. By examining these counterfactual scenarios, we gain a deeper understanding of the factors influencing the model's decision-making process and the conditions under which different outcomes might occur.

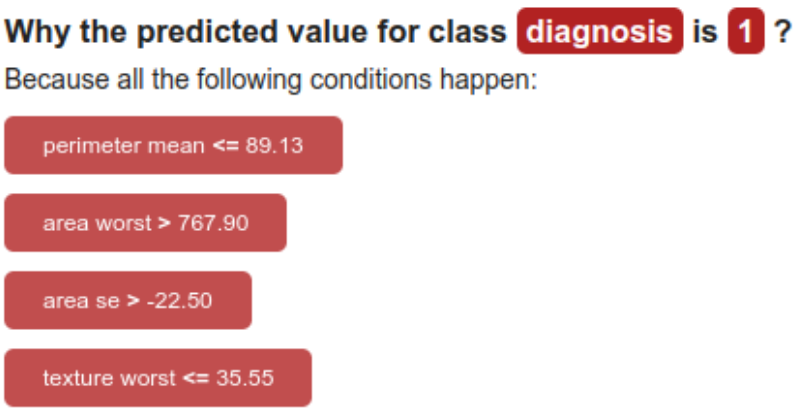


Figure 21: **Lore explanation of rules contributed in classification for third instance**

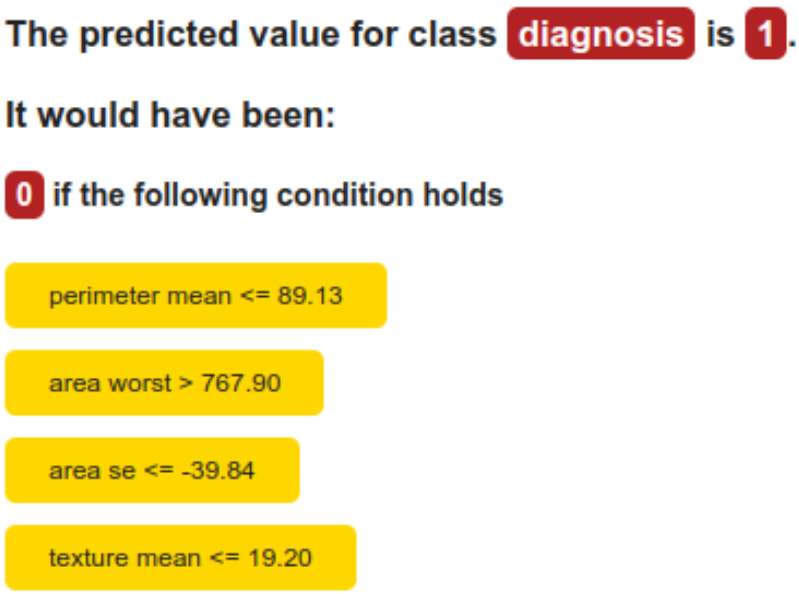


Figure 22: **Plot of counter factual Rules for third instance**

In the second misclassification instance, the utilized model classified the instance as belonging to class 1, indicating a diagnosis of malignancy. According to the LORE

explanation presented in Figure 21, this classification was influenced by specific feature values. Specifically, the perimeter mean was observed to be under 89.13, and the texture worst was lower than 35.55. These particular values of the features played significant roles in driving the classification towards class 1, indicating malignancy.

Furthermore, the counterfactual rules presented in the explanation illustrate conditions under which the classification outcome would have been opposite. For instance, if the texture mean had been lower than 19.2, the model would have classified the instance differently. This insight provides valuable information about the sensitivity of the model to variations in feature values and highlights potential scenarios where different outcomes might occur.

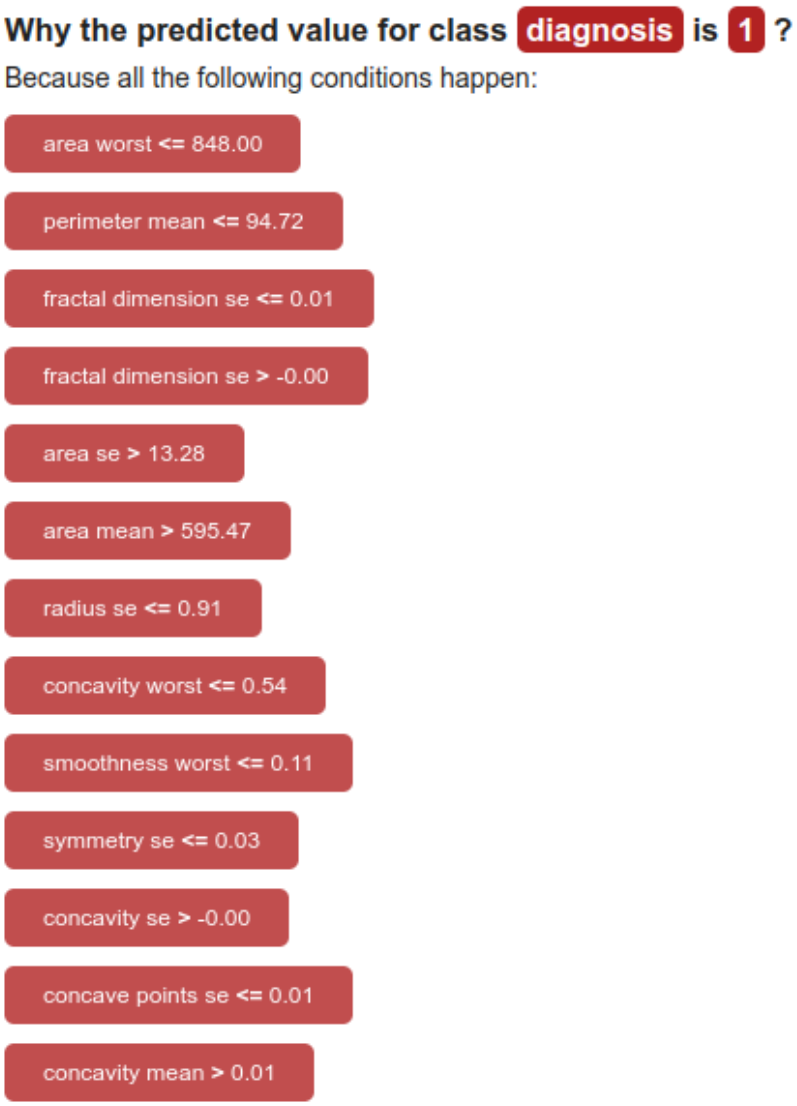


Figure 23: **Lore explanation of rules contributed in classification for third instance**

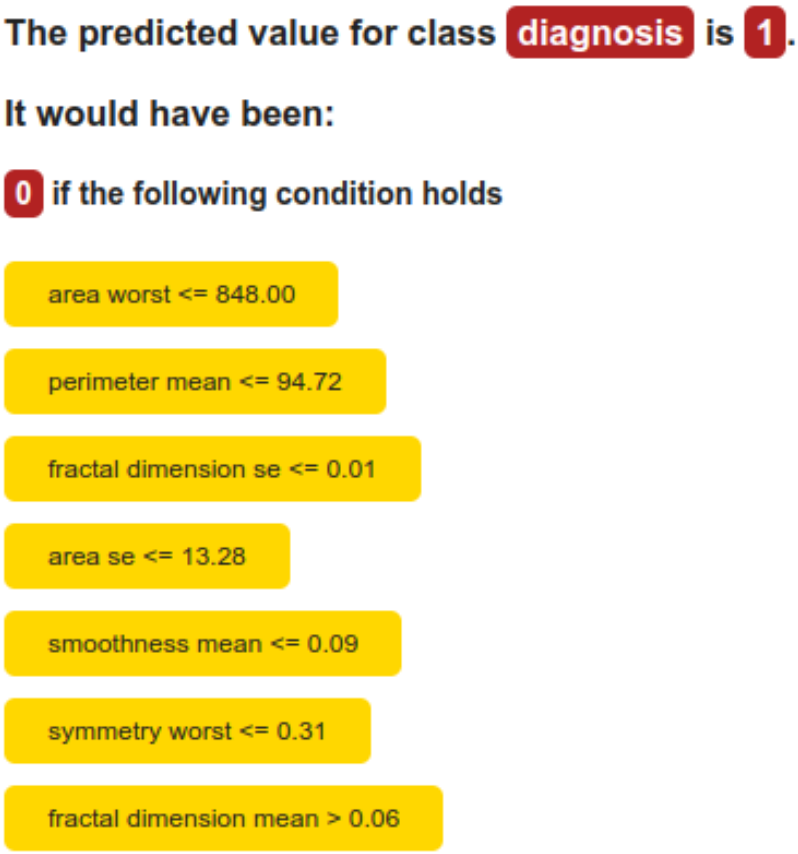


Figure 24: **Plot of counter factual Rules for third instance**

In the third instance of misclassification, the model categorized the instance as belonging to class 1. Figure 23 provides insights into the features that made significant contributions to this decision. These features likely played crucial roles in driving the classification towards class 1.

Additionally, the counterfactual plot presented alongside the explanation illustrates potential scenarios where the classification outcome would have changed. Specifically, changes in features such as area-se, smoothness mean, symmetry-worst, and fractal dimension mean could lead to a different classification outcome, resulting in the instance being classified as class 0 instead.

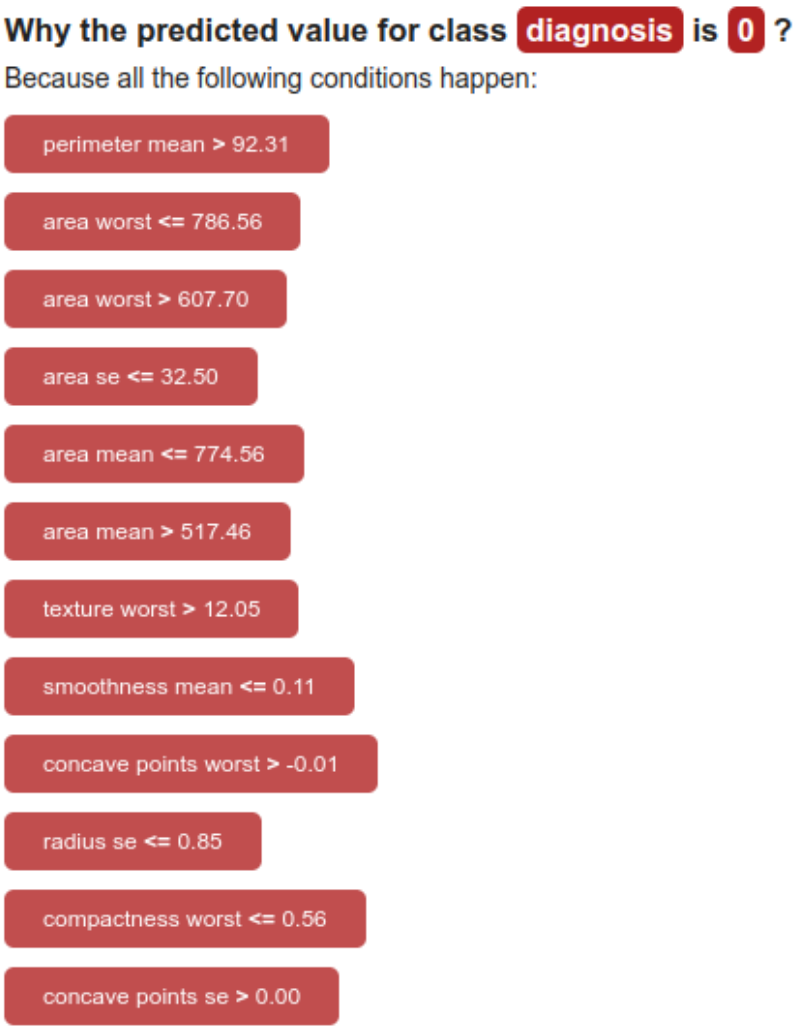


Figure 25: **Lore explanation of rules contributed in classification for first instance of low confidence**

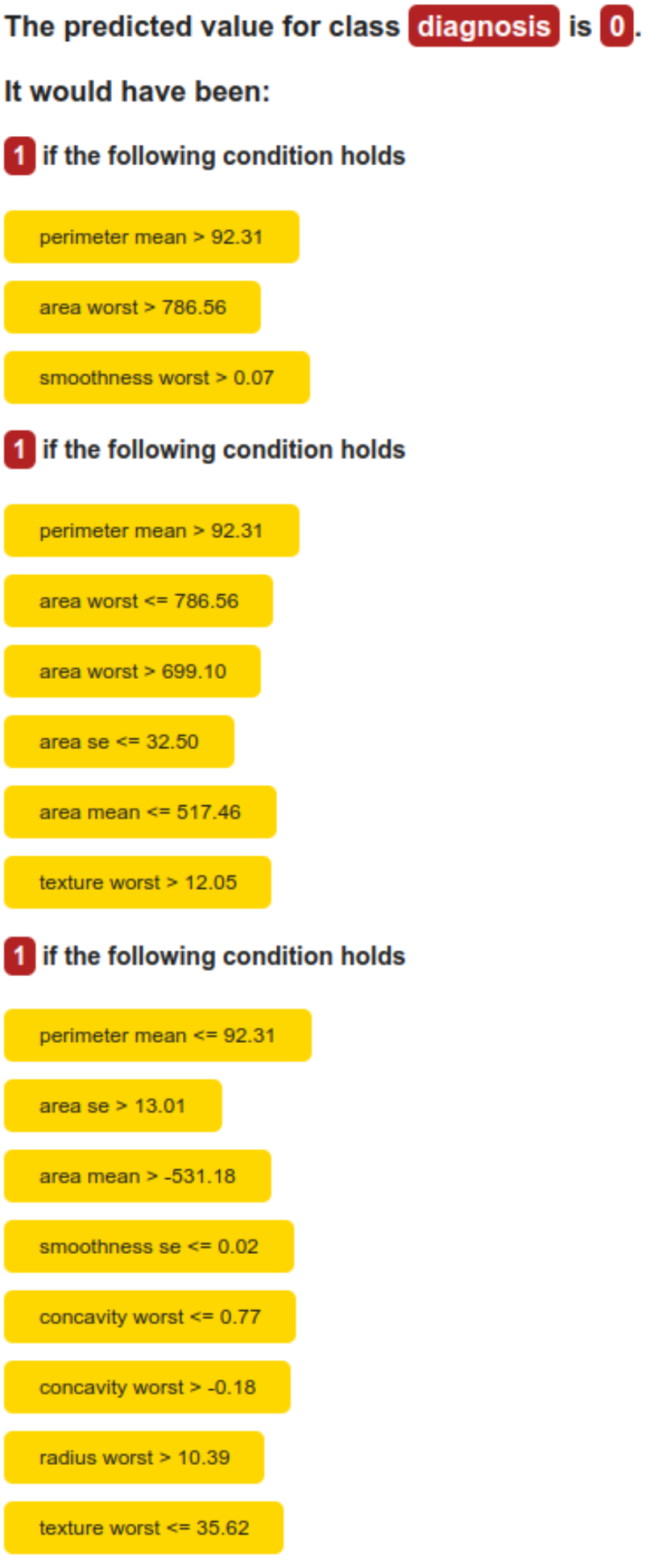


Figure 26: **Plot of counter factual Rules for first instance of low confidence**

In the first instance of the low confidence category, our model classified the data as benign (class 0). As illustrated in Figure 25, specific features such as smoothness mean lower than 0.11 and radius se lower than 0.85 were among the instance features that contributed to the model's decision. These features likely played a significant role in guiding the model towards classifying the instance as benign.

Furthermore, the figure 26 demonstrates that changes in certain features could potentially affect the model's outcome. Features such as smoothness worst, area worst, and area mean are highlighted as examples.

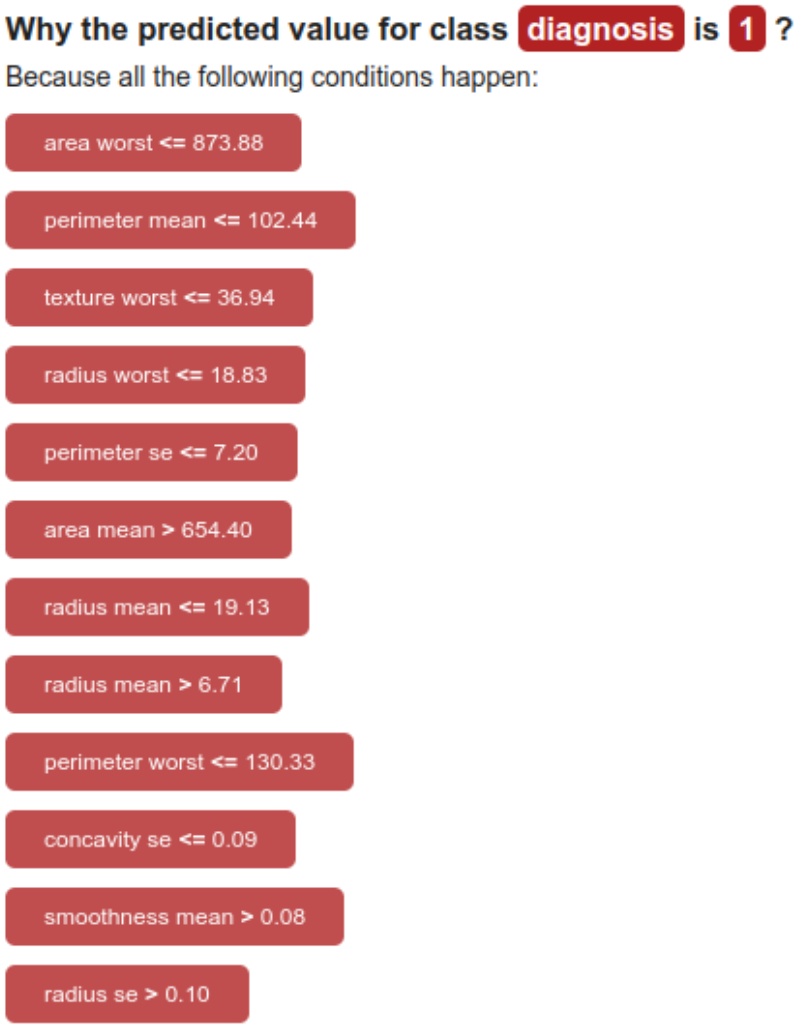


Figure 27: **Lore explanation of rules contributed in classification for second instance of low confidence**

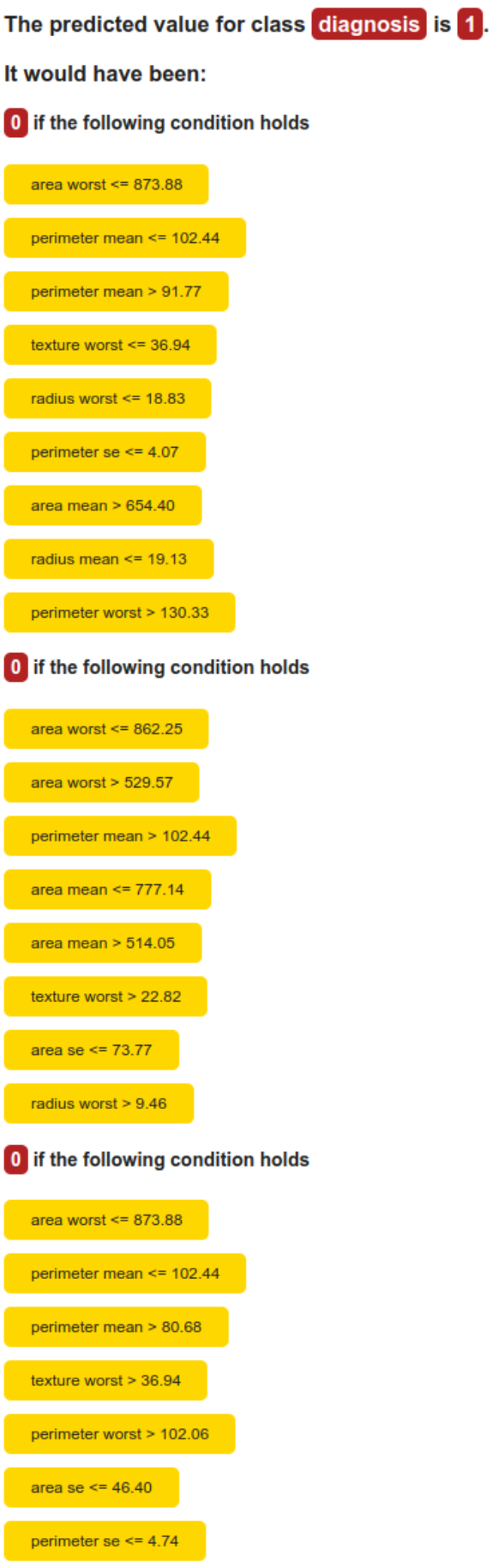


Figure 28: **Plot of counter factual Rules for second instance of low confidence**

In the second instance of the low confidence category, our model predicted the data as belonging to class 1. Figure 27 illustrates that the area-worst feature and perimeter-mean feature played crucial roles in this decision. These features likely contributed significantly to the model's classification of the instance as class 1.

Conversely, the figure also indicates that the classification would have favored class 0 under certain conditions. For example, if features such as perimeter se were lower than 4.07 and radius mean were lower than 19.13, the model would have classified the instance differently, likely as class 0.

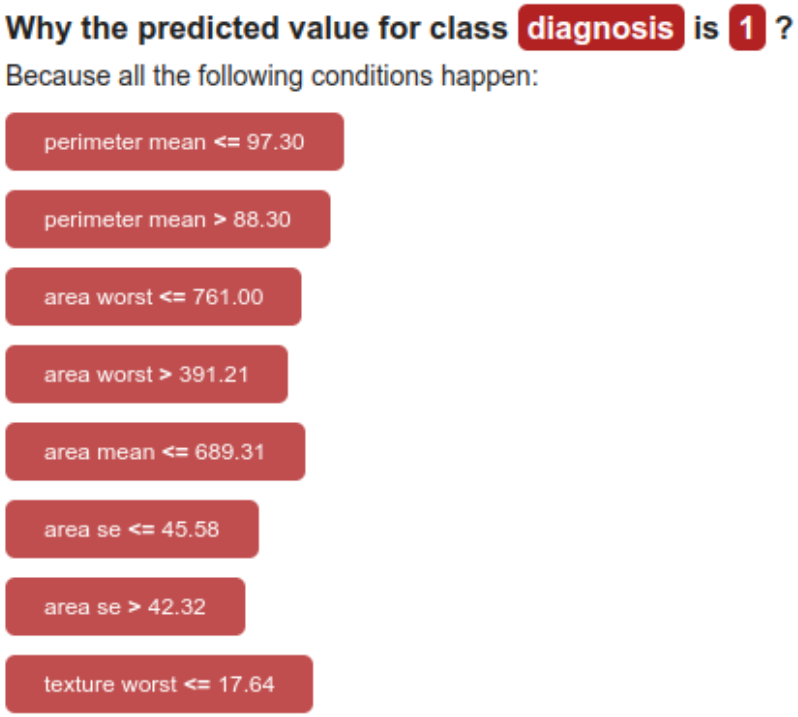


Figure 29: **Lore explanation of rules contributed in classification for third instance of low confidence**



Figure 30: **Plot of counter factual Rules for third instance of low confidence**

0.3 SHAP

SHAP (SHapley Additive exPlanations) is a technique used to elucidate the individual predictions generated by machine learning models. It achieves this by breaking down each prediction into a sum of contributions from every input variable of the model. This method allows for a nuanced understanding of how each feature influences the model's output. Additionally, SHAP values vary across different instances in the dataset, capturing the unique impact of each feature for every data point. The output of SHAP is a dataset mirroring the original input dimensions, but instead of containing raw data, it holds SHAP values. These values, along with a fixed base value, can replicate the model's predictions for each instance. For regression models, the base value corresponds to the mean of the target variable, while for classification models, it reflects the prevalence of the positive class. By analyzing SHAP values, insights emerge regarding the intricate relationship between input variables and model predictions, both on a per-instance basis and across the entire dataset. To avoid exhaustive reporting, we will interpret one instance from each category. Figure 31 (Beeswarm plot) is a more complex and information-rich display of SHAP values that reveal not just the relative importance of features, but their actual relationships with the predicted outcome.The horizontal axis indicates how predictive that feature is for that individual in a positive (= positive label) or negative direction. In addition, the beeswarm chart adds color coding of each dot based on how the feature value for that individual compares to the average for the entire population. Each instance the given explanation is represented by a single dot on each feature fow. The x position of the dot is determined by the SHAP value of that feature, and dots "pile up" along each feature row to show density. Color is used to display the original value of a feature. According to Figure 31, we can say area-worst, area-mean and area-se are important features for model to classify instants to class 0 or 1. Moreover, area-worst and area-mean features illustrates that almost all of the instants classified as class 1, value for these features are over mean SHAP value. Also, perimeter-mean and texture-worst features have opposite behaviour, as they have lower values it is more likely for instance to be malignant.

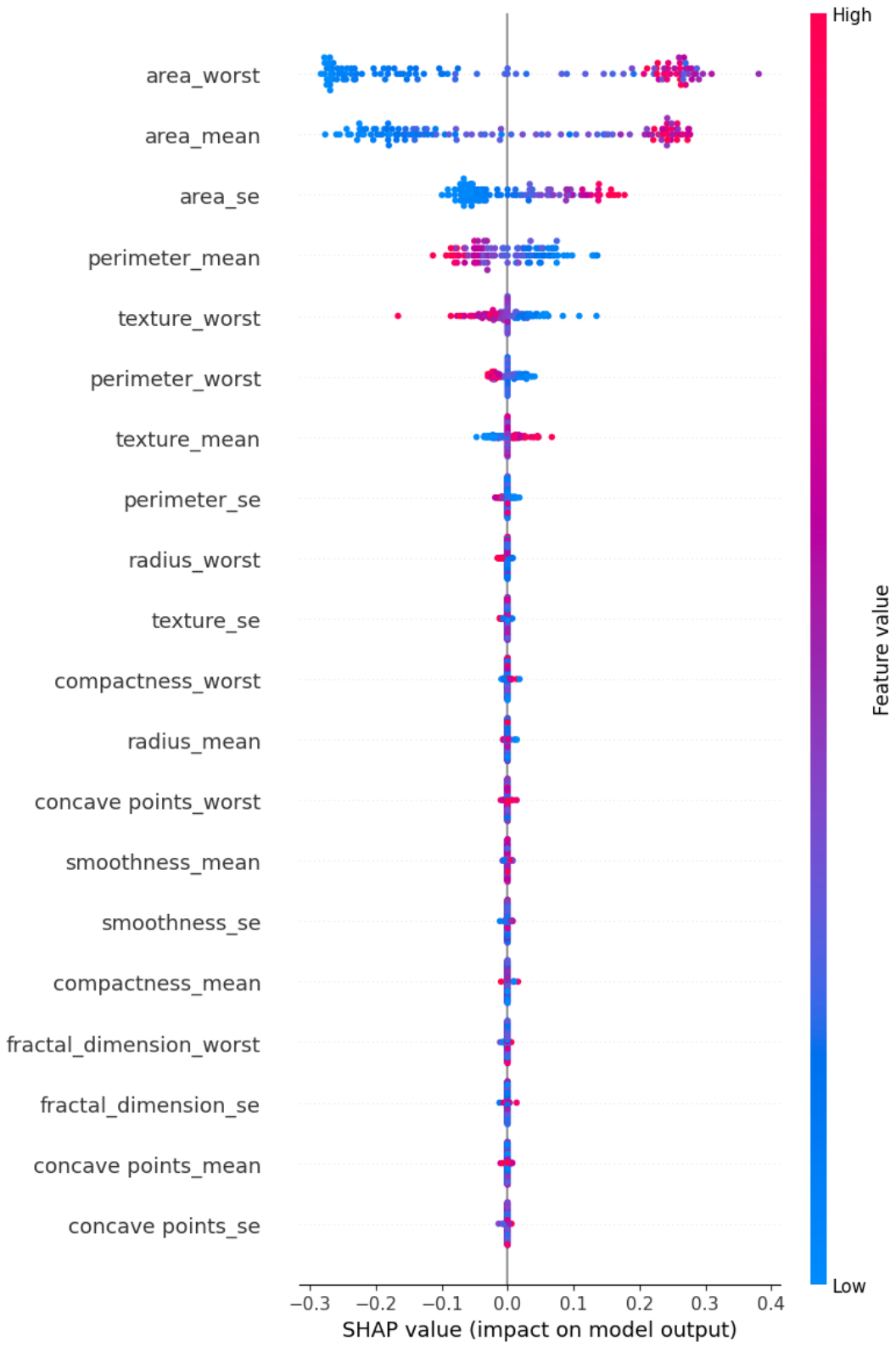


Figure 31: **Beeswarm plots of SHAP Explanation.**

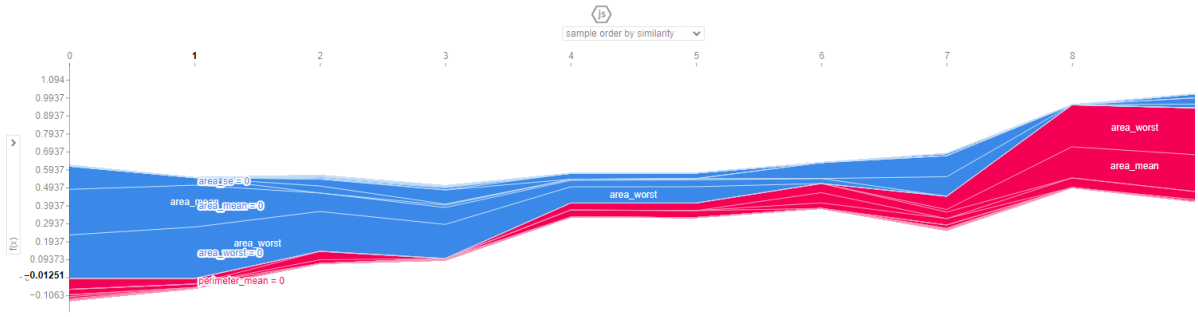


Figure 32: **Stacked Force plot of SHAP for all ten instances.**

The waterfall plot provides a comprehensive breakdown of a single prediction, offering insights into the contributions of each feature towards the probability of a specific outcome, such as being malignant or benign, within the dataset. Whereas waterfall plots are expansive and spare no detail when explaining a prediction, force plots are equivalent representations that display the key information in a more condensed format (Figure 34). Illustrated in Figure 33, this example showcases how each feature influences the model's prediction, emphasizing their additive nature as positive and negative contributors. The structure of the waterfall plot highlights how these contributions, along with the base value, collectively determine the model's prediction, denoted as f(x). For instance, in Figure 33, the model's prediction for the third instance of misclassification is depicted as 14%. This value represents the sum of the base value, which accounts for 49% of the prediction, along with the sum of SHAP values depicted by the arrows. Notably, the contributions of features such as area-worst and area-mean are emphasized in the plot, indicating their significant impact on the model's outcome. This observation is further corroborated by the force plot in Figure 34, where the dominance of blue values underscores their pronounced effect on the model's prediction.

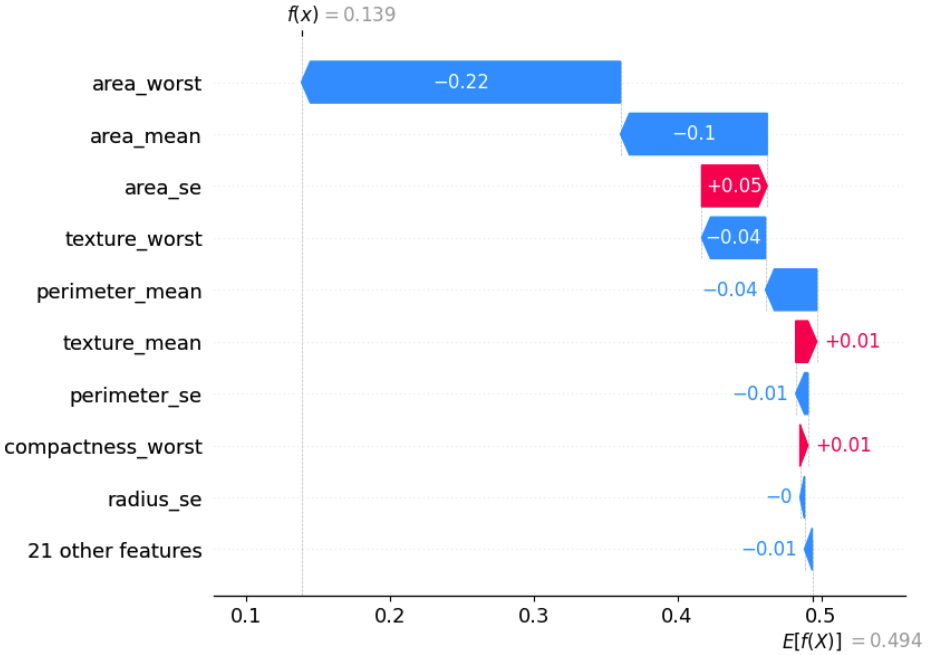


Figure 33: **Waterfall plot of SHAP for third instance of misclassification category.**

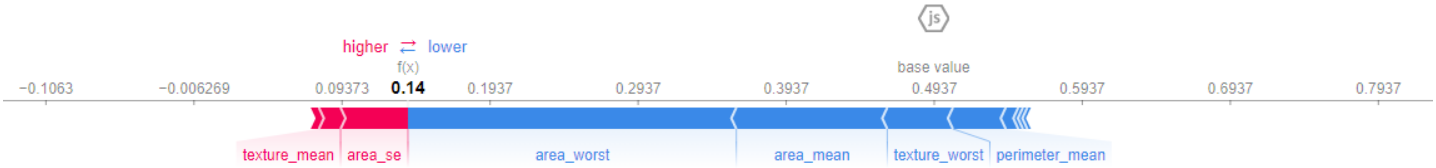


Figure 34: **Force plot of SHAP for third instance of misclassification category.**

Furthermore, the utilized model accurately predicted this instance as class 1 with minimal deviation. As depicted in Figure 33, the model's average base value, denoted as E[f(x)], stands at 49%. Further insight from Figures 35 and 36 reveals the intricate dynamics of feature contributions. Specifically, texture-worst, area-mean, and perimeter-se features emerge as influential factors favoring class 0. Conversely, features such as area-se, perimeter-mean, texture-mean, and compactness-worst contribute to the model's prediction of class 1. These observations underscore the complex interplay of features in the model's decision-making process.

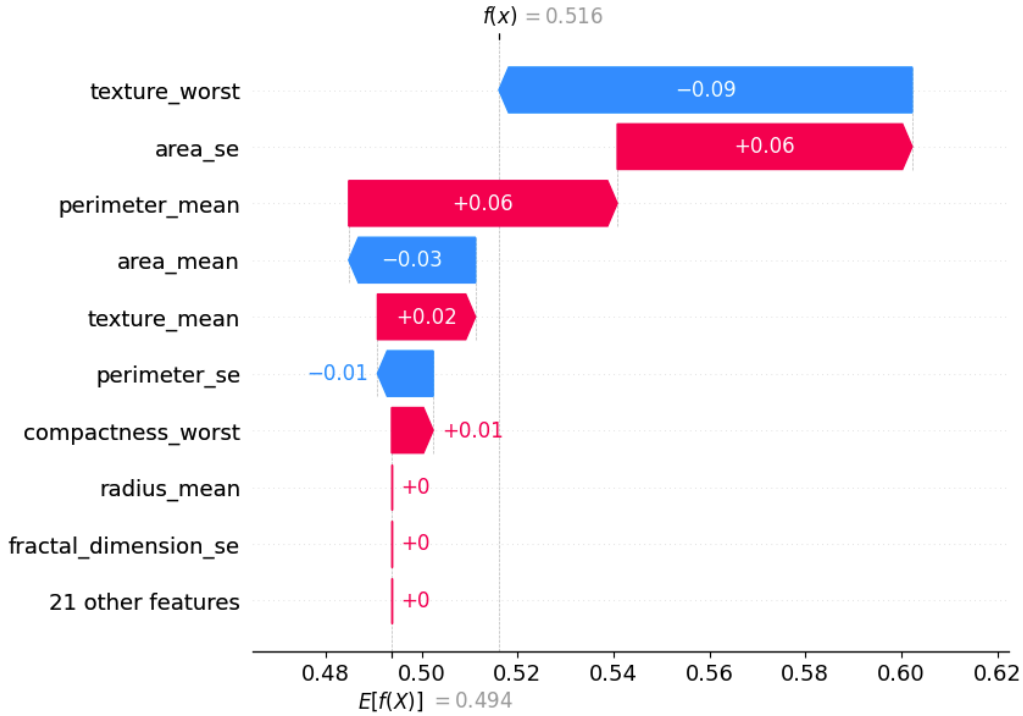


Figure 35: **Waterfall plot of SHAP for first instance of Low confidence category.**

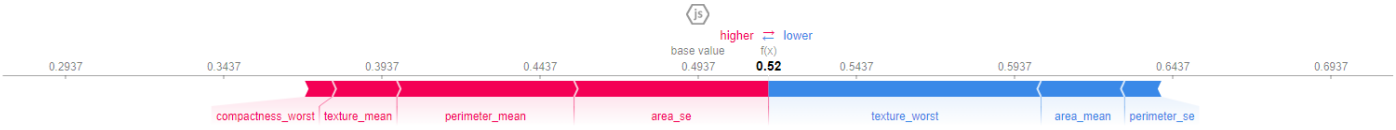


Figure 36: **Force plot of SHAP for first instance of Low confidence category.**

Lastly, within the High confidence category, the model accurately predicted the true class with a high probability. According to the model's prediction, this patient was labeled as class 0. Illustrated by the SHAP values in Figure 37, it's evident that features

such as area-worst and area-mean play pivotal roles in determining this outcome, with respective contributions of 0.27 and 0.21. Furthermore, the force plot generated by the SHAP method in Figure 38 corroborates these findings, highlighting the significant contribution of these features. Consequently, the model's prediction of class 0 is reinforced, achieving a confidence level of 94%.

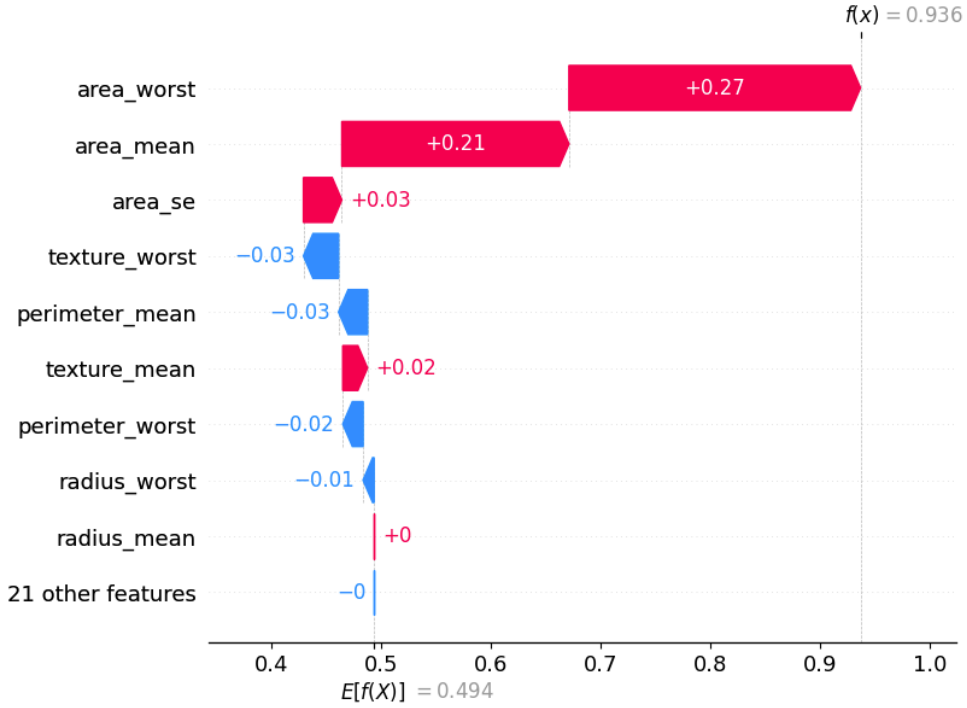


Figure 37: **Waterfall plot of SHAP for third instance of High confidence category.**

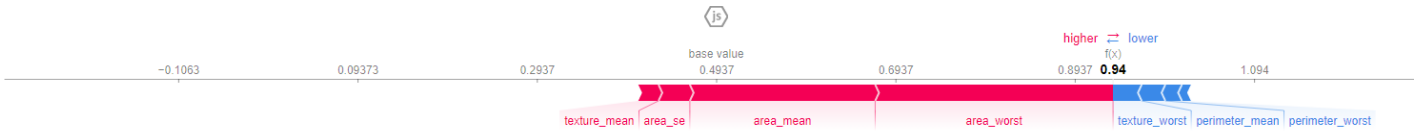


Figure 38: **Force plot of SHAP for third instance of High confidence category.**

Interpretable Models

```
Coefficients (log odds): [[-1.18729907 -0.51567116 -0.13739033  0.01339158  0.04505652  0.21584064
   0.29831245  0.12752774  0.06724706  0.01360227 -0.04083591 -0.51104543
  -0.04311381  0.09366359  0.00420175  0.04434217  0.06114392  0.01641796
   0.01584365  0.00388621 -1.24856219  0.54821564  0.20627851  0.01460484
   0.08028862  0.65250215  0.8065715   0.23606627  0.21690699  0.06055259]]
```

Figure 39: **Coefficients of the Linear Regression Model.**

In the last part of the project, we trained a linear regression model on the tumor dataset using Python package sklearn. After training the model, we obtained 30 coefficients of the model (Figure 39. Coefficients (also known as log odds ratios) are the values that

multiply the predictor values. Then, we predicted the class for 10 instances using the trained LR model. The model predicted that only instances 7 and 8 belong to Class 1 (malignant tumor diagnosis), while other instances belong to Class 0 (benign tumor diagnosis). In order to see the confidence of these predictions, we calculated probability estimates for for both classes in each instance (shown on titles of Figures **??-??**. Results show high confidence between 89%-100% (the lowest confidence is for Instance 1).

Figure 40 contains bar plots which show how each feature contributed to the predicted class of each instance chosen. These figures tell us that features have similar contributions for those 10 instances. Features 3 (*area_mean*), 21 (*texture_worst*), 22 (*perimeter_worst*) and 23 (*area_worst*) have significantly positive contribution, while features 0 (*radius_mean*), 1 (*texture_mean*), 2 (*perimeter_mean*) and 20 (*radius_worst*) have negative contribution. Other features have either small or no contribution. From these contributions, we can say that maximum measures of the tumor area, texture and perimeter contribute to the prediction positively (increasing the probability of the malign tumor), where mean area is also significant. On the other hand, mean measures of the radius, texture and perimeter have negative contributions (increasing the chance of benign tumor), maximum radius having significant contribution, too. However, there are small differences in plots across instances which come from various contribution to the log-odds of aforementioned features.

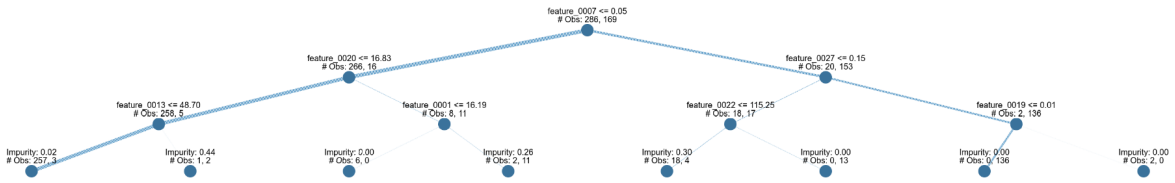
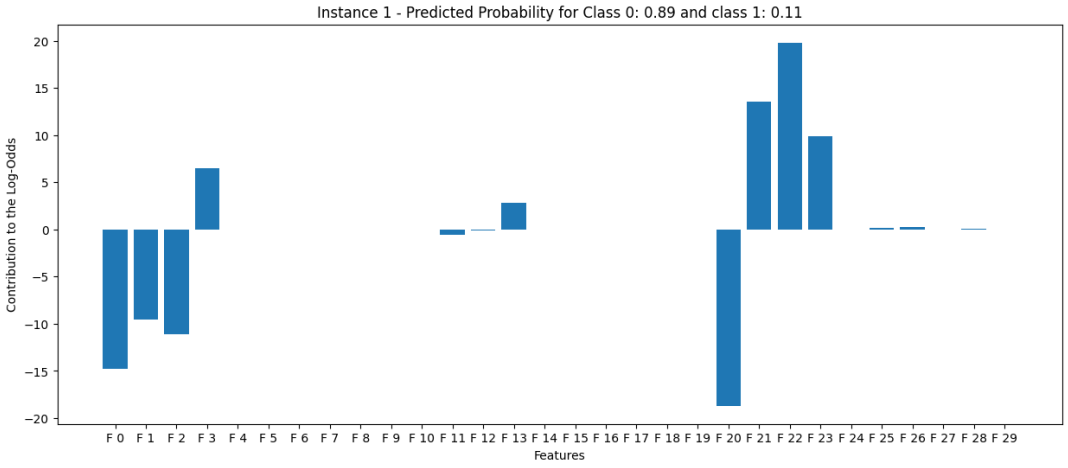
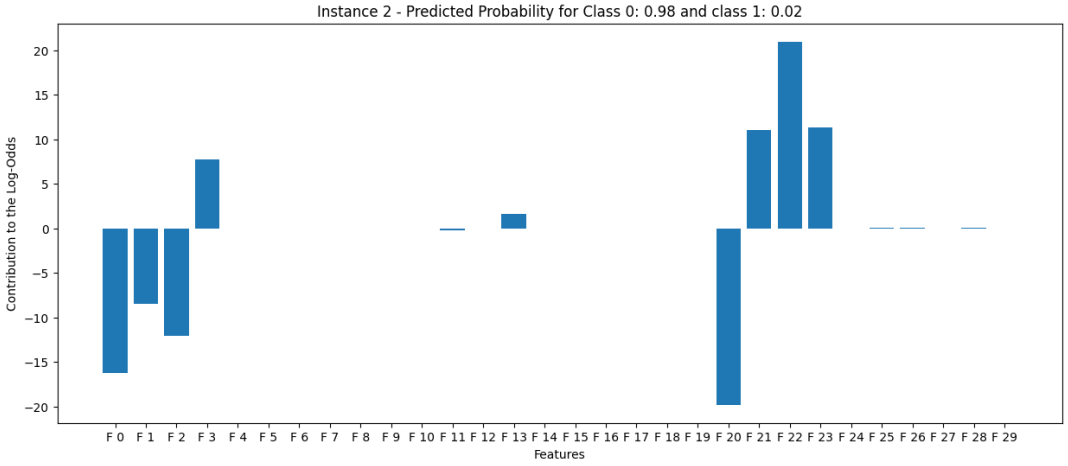


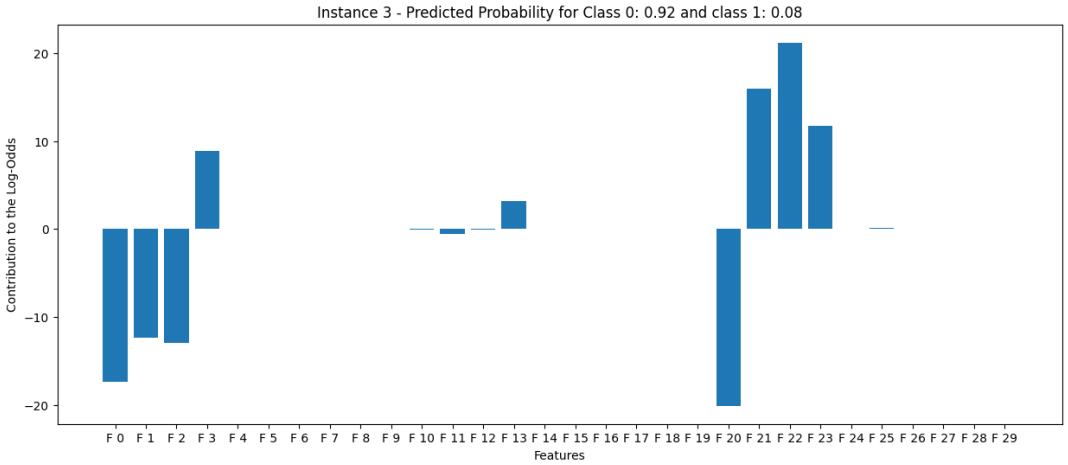
Figure 41: **Decision Tree of the Explainable Model.**

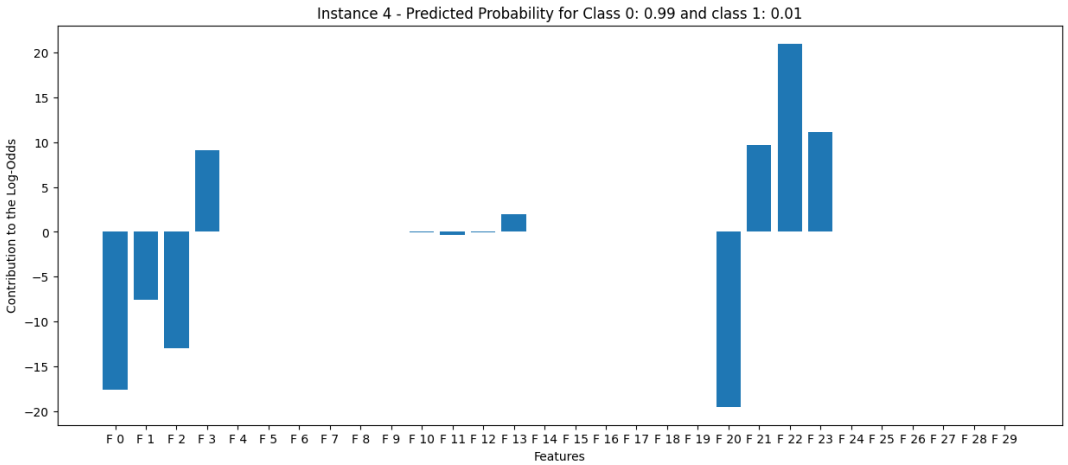
Another way to explain an interpretable model is to use classification trees. We used a Python package called interpretML to solve this task. This package produces interactive plots, where we can see the general (global), or instance-based (local) behavior of the model. In Figure 41, the overall decision process of the model shows two major paths: one going to the left which classifies the instance as benign, and another one which goes to the right and classifies the tumor as malignant. The first feature in classification is feature 7 (*concanve points_mean*), which was not observed in previous analyses. Other features considered to identify a benign tumor are feature 20 (*radius_worst*) and 13 (*area_se*), while features 27 (*concave points_worst*) and 19 (*fractal_dimension_se*) are main features used to identify a malign tumor. This tree shows that identification of a benign tumor uses more familiar features used in previous analyses.

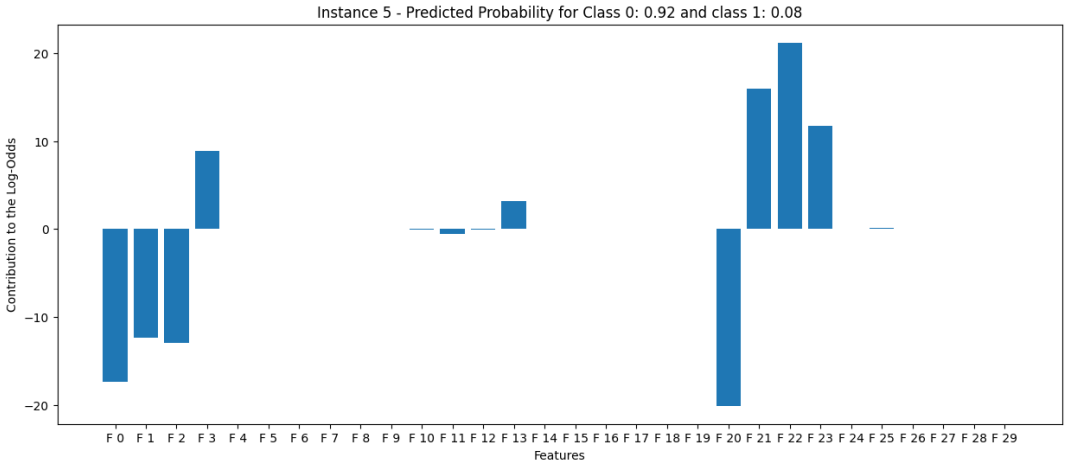
We also used this decision tree to check the decision process in 10 instances chosen, which is shown in Figure 42. There are two instances with misclassified labels (classification trees 22 and 23), both of which were predicted by a neural network with low confidence. The first instance has all features which fit a benign tumor, but it is a malignant tumor (there are only 3 such cases versus 257). The second misclassified instance has a more interesting path in the decision tree: the value in feature 7 fits a malignant tumor, but a low value in feature 27 takes the path to the feature 22 (*perimeter_worst*). Here, there is a low confidence to take a decision (18 versus 17 labels), this is why the model splits the subset based on Gini impurity, where the misclassified observation falls

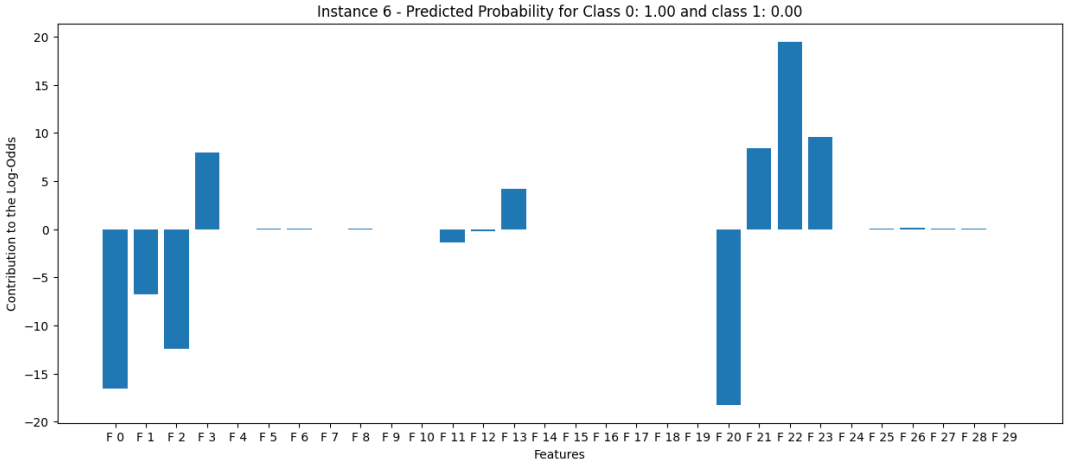

Instance 1 - Predicted Probability for Class 0: 0.89 and class 1: 0.11
Contribution to the Log-Odds
Features

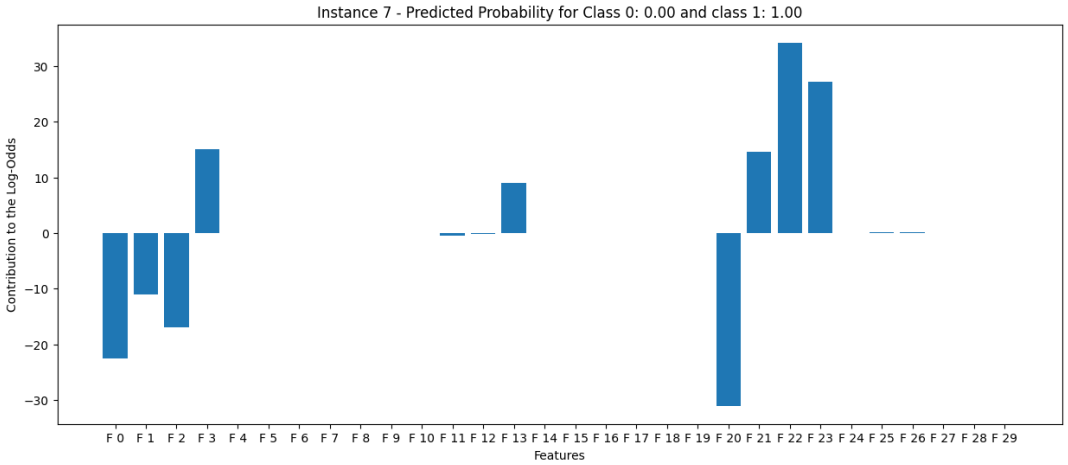

Instance 2 - Predicted Probability for Class 0: 0.98 and class 1: 0.02
Contribution to the Log-Odds
Features

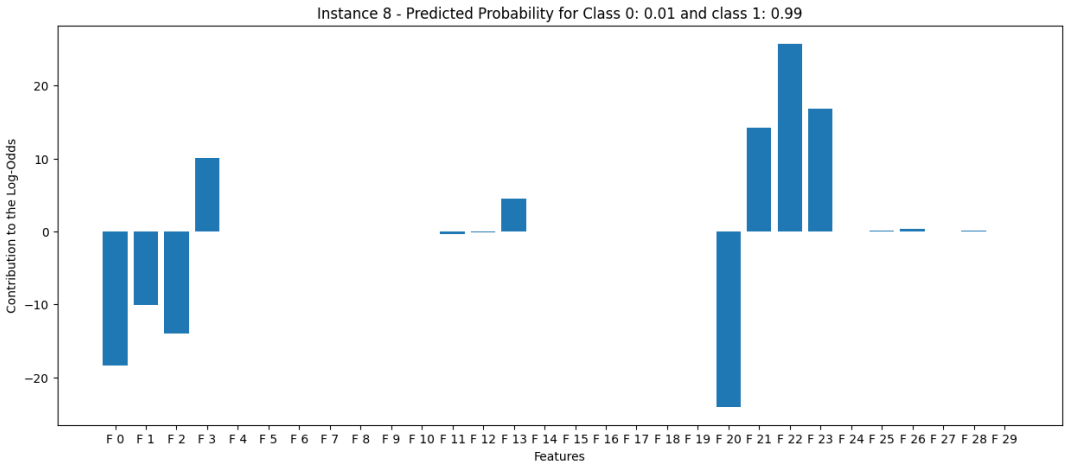

Instance 3 - Predicted Probability for Class 0: 0.92 and class 1: 0.08
Contribution to the Log-Odds
Features


Instance 4 - Predicted Probability for Class 0: 0.99 and class 1: 0.01
Contribution to the Log-Odds
Features


Instance 5 - Predicted Probability for Class 0: 0.92 and class 1: 0.08
Contribution to the Log-Odds
Features


Instance 6 - Predicted Probability for Class 0: 1.00 and class 1: 0.00
Contribution to the Log-Odds
Features


Instance 7 - Predicted Probability for Class 0: 0.00 and class 1: 1.00
Contribution to the Log-Odds
Features


Instance 8 - Predicted Probability for Class 0: 0.01 and class 1: 0.99
Contribution to the Log-Odds
Features

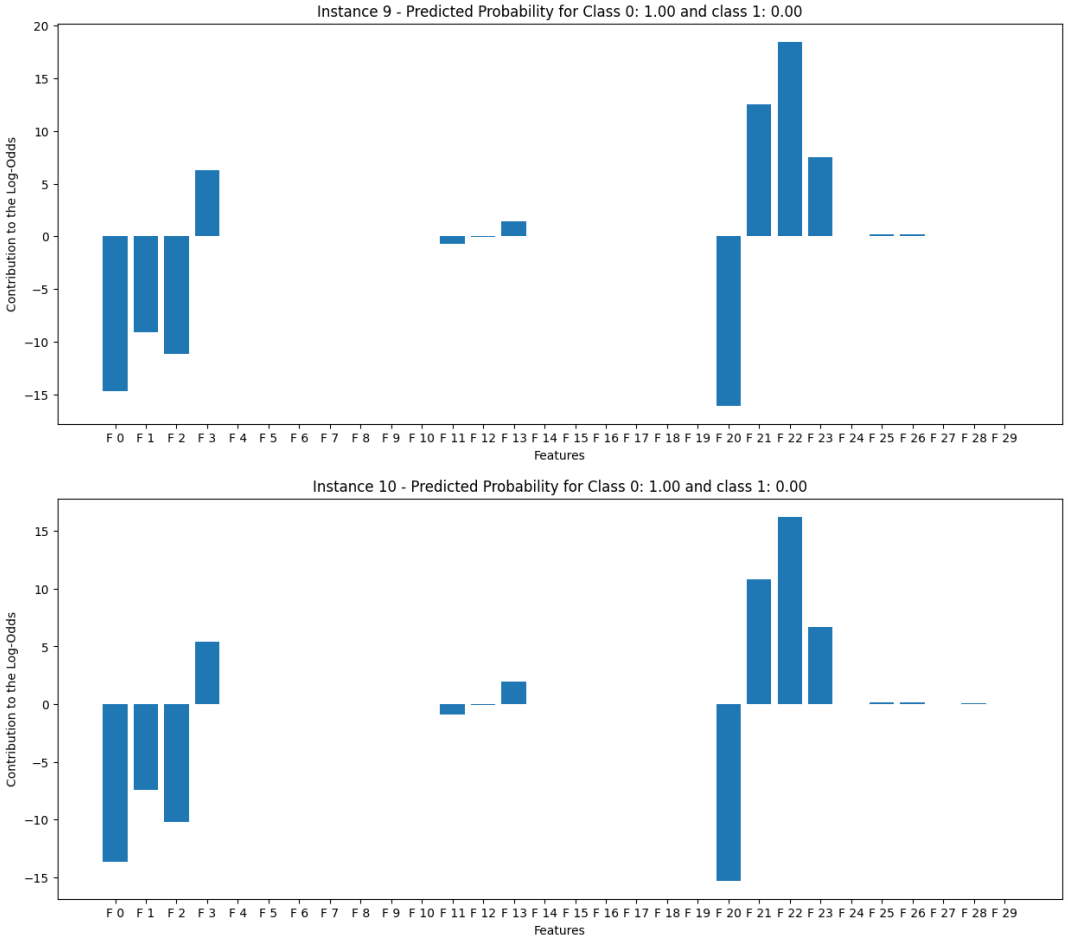
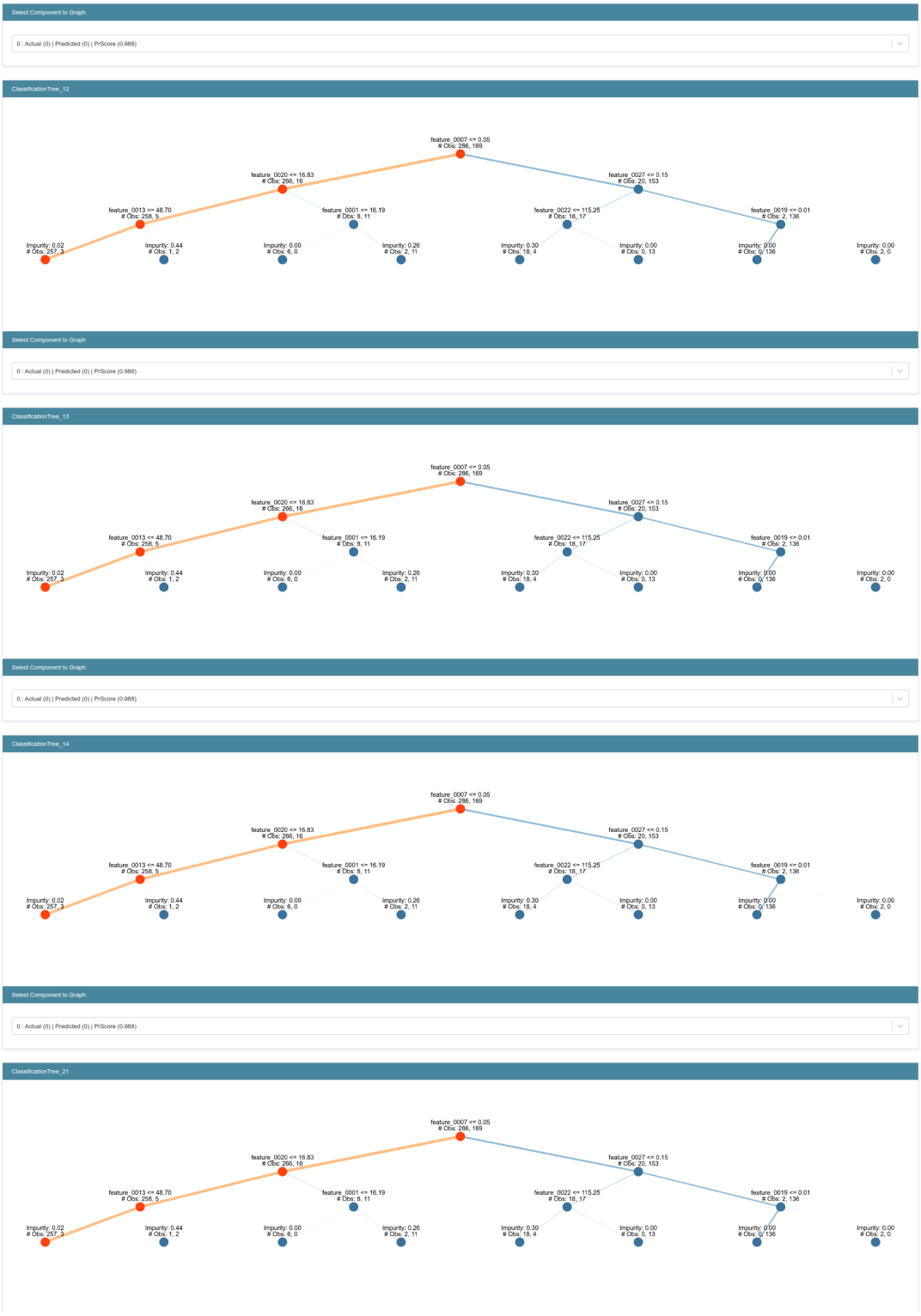
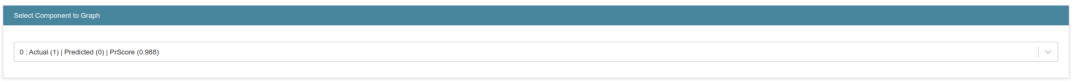
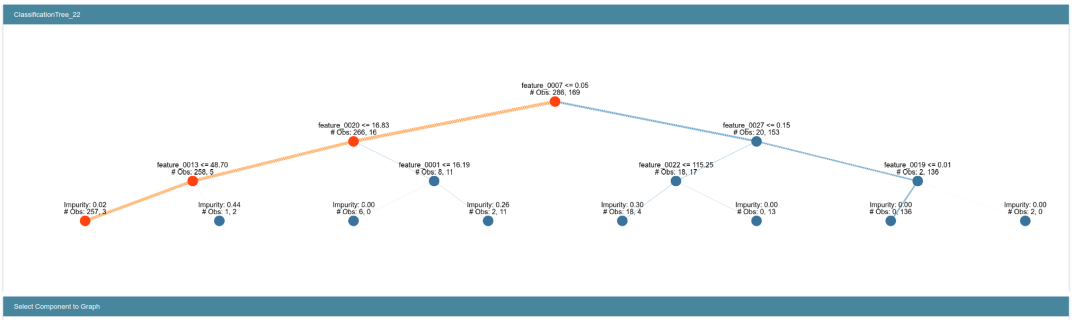
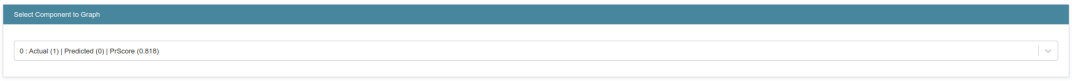


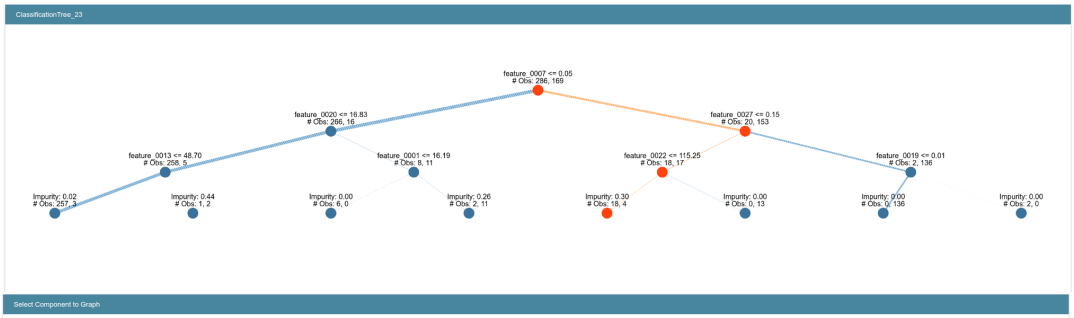
Figure 40: **Contribution of Features in 10 Instances Chosen**

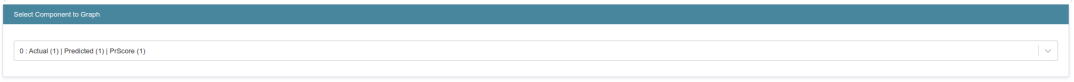

Select Component to Graph
0 : Actual (0) | Predicted (0) | PrScore (0.988)
ClassificationTree_12
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0
Select Component to Graph
0 : Actual (0) | Predicted (0) | PrScore (0.988)
ClassificationTree_13
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0
Select Component to Graph
0 : Actual (0) | Predicted (0) | PrScore (0.988)
ClassificationTree_14
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0
Select Component to Graph
0 : Actual (0) | Predicted (0) | PrScore (0.988)
ClassificationTree_21
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0

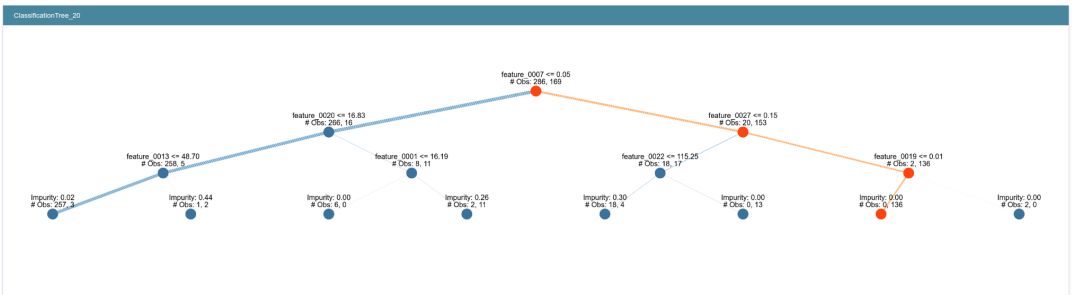

Select Component to Graph
0 : Actual (1) | Predicted (0) | PrScore (0.988)

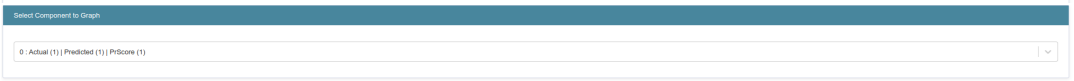

ClassificationTree_22
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0

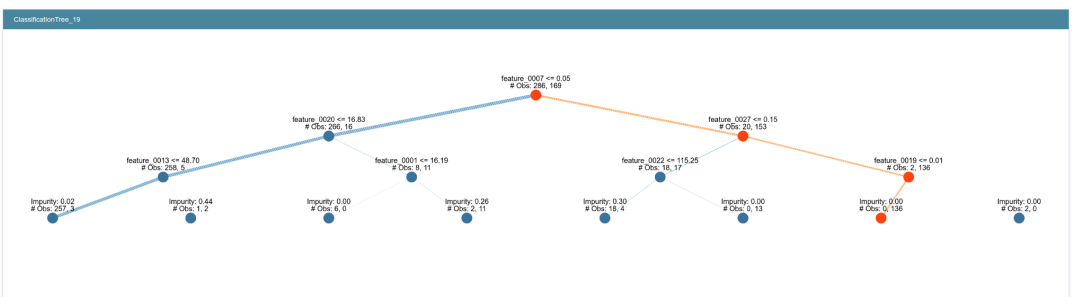

Select Component to Graph
0 : Actual (1) | Predicted (0) | PrScore (0.818)


ClassificationTree_23
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0


Select Component to Graph
0 : Actual (1) | Predicted (1) | PrScore (1)


ClassificationTree_20
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0


Select Component to Graph
0 : Actual (1) | Predicted (1) | PrScore (1)


ClassificationTree_19
feature_0007 <= 0.05
Obs: 286, 169
feature_0020 <= 16.83
Obs: 266, 16
feature_0027 <= 0.15
Obs: 20, 153
feature_0013 <= 48.70
Obs: 258, 5
feature_0001 <= 16.19
Obs: 8, 11
feature_0022 <= 115.25
Obs: 18, 17
feature_0019 <= 0.01
Obs: 2, 136
Impurity: 0.02
Obs: 257, 3
Impurity: 0.44
Obs: 1, 2
Impurity: 0.00
Obs: 6, 0
Impurity: 0.26
Obs: 2, 11
Impurity: 0.30
Obs: 18, 4
Impurity: 0.00
Obs: 0, 13
Impurity: 0.00
Obs: 0, 136
Impurity: 0.00
Obs: 2, 0

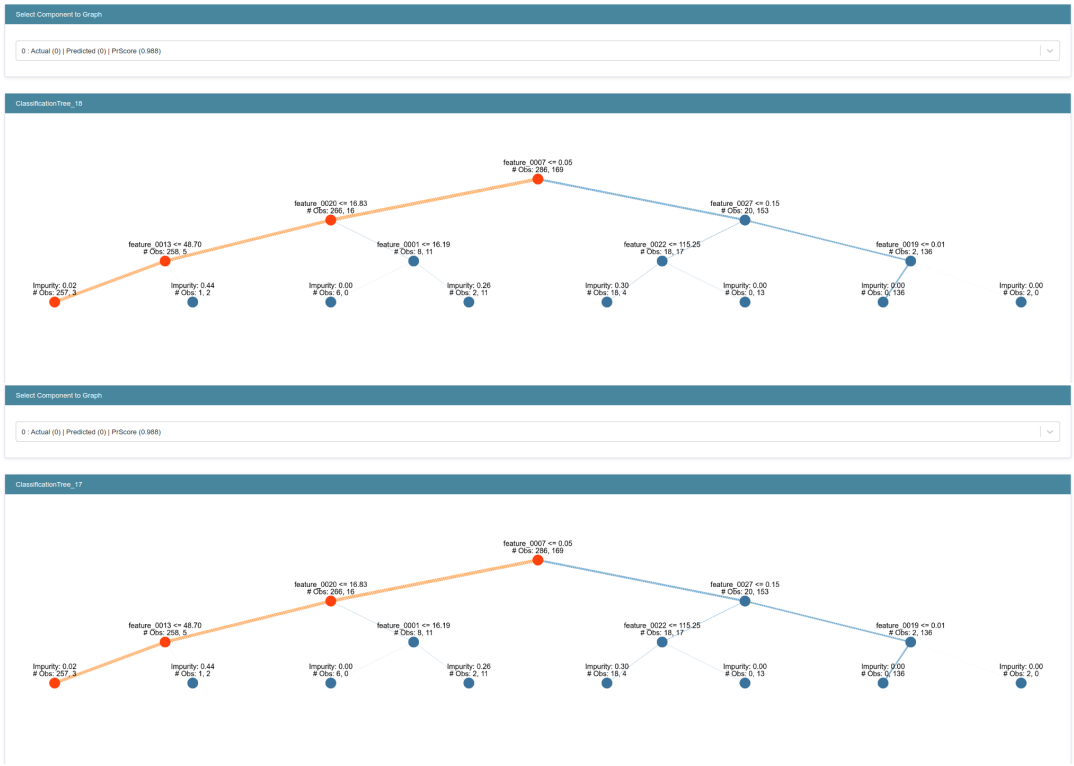


Figure 42: **Decision Trees for 10 Instances Chosen**

into (18 versus 4 labels).

In the end, we also evaluated the fidelity of the interpretable model, where we compared the prediction of the black box with the explainer on 10 instances. The accuracy was 50%.

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